

Theme: Urban city

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By the end of the theme, students will be able to:

- Identify and name key technologies present in smart cities.
- Describe how smart cities use resources wisely, keep things clean, and use less energy to help the Earth.
- Recognize and make basic two-dimensional shapes (e.g., circles, squares) and three-dimensional shapes (e.g., cubes, spheres).
- Recognize the Tinkercad interface and become familiar with its basic tools and functions.
- Compare and contrast the vertices, edges, and faces of different three-dimensional shapes.
- Recognize basic road safety rules and understand how they help keep us safe while traveling.
- Identify forces and explore simple machines, like ramps, using measuring tools to understand dimensions.
- Understand the basics of electricity and create simple circuits.
- Recognize and understand the basics of algorithmic thinking by following simple step-by-step instructions to solve problems.
- Evaluate the effectiveness of traffic light models in demonstrating how to solve road accidents at crossroads and smart alarms in cross-sections.

UNIT 01

Chapter 1: Design

Lesson 01

Building smart city

Objectives:

- 1-**Identify common traffic dangers** in cities and discuss why safety is important for both **pedestrians and drivers**.
- 2-**Recognize basic traffic signs** and explain their meanings.
- 3-**Understand safe crossing behaviors** using the "Stop, Look, Listen" rule.
- 4-**Discuss the importance of accessibility** in cities and how to make public spaces safer for **everyone, including people with disabilities**.
- 5-**Brainstorm solutions** to improve city safety and express their ideas through a **poster activity**.

Challenge Questions of the Lesson

- 1-What would you change in your city to make it safer for everyone?
- 2- Why is it important to have clear traffic signs and signals?
- 3- How can we teach others to follow safety rules in the city?

SEL relation

1. **Self-Awareness** – Recognizing personal feelings about city safety and the importance of following rules.
2. **Self-Management** – Practicing patience and making safe decisions while walking in the city.
3. **Social Awareness** – Understanding the needs of others, like blind individuals, and promoting empathy.
4. **Relationship Skills** – Working together on activities and communicating safety reminders.
5. **Responsible Decision-Making** – Thinking critically about safe choices and improving city safety.

1. Engage

Teacher Role:

- Introduce the story scene by setting the stage with an engaging discussion:
 - "Imagine you're on a school bus with your friend, and suddenly, something unexpected happens! How would you feel?"
- Use storytelling techniques (animated voice, gestures, visuals) to bring Adam and Laila's experience to life.
- Show relevant visuals (a school bus, a busy road, a traffic light) to help students visualize the setting.
- Ask thought-provoking, open-ended questions to spark curiosity:
 - "Why do accidents happen in cities?"
 - "What could help make roads safer?"
- Encourage students to **predict** what Adam might do next based on the story.
- Foster a supportive classroom environment where students feel comfortable sharing their ideas.

Expected From Students:

- Actively listen and participate in the discussion by sharing their thoughts and asking questions.
- Engage with visuals and storytelling elements to deepen their understanding of the scene.
- Make predictions about the story based on clues from the discussion and visuals.
- Relate the story to their own experiences (e.g., times they've seen traffic or road safety issues).
- Begin thinking critically about real-world problems and how technology could help solve them.

1. Preview with a Hook

Strategy: Begin with an intriguing question or striking image.

Implementation:

- Display an image of a **busy intersection with traffic lights and cars** versus an **empty, quiet road**.
- Ask: “**Which road looks safer? Why?**”
- Briefly discuss students’ thoughts to introduce the idea of road safety.

2. Make Predictions

Strategy: Use the story’s title and an image to spark curiosity.

Implementation:

- Show the story title: “**Adam and Laila’s School Bus Adventure.**”
- Ask: “**What do you think happens in the story?**”
- Have students share their ideas and revisit their predictions later in the lesson.

3. Set the Scene with a Visual or Sound

Strategy: Use visuals and sounds to immerse students in the story.

Implementation:

- Show a picture of a **school bus and a city street**.
- Play a short **sound clip of traffic noises** (honking, sirens, crosswalk beeps).
- Ask: “**What sounds do you hear in a city? How do they help people stay safe?**”

4. Act It Out

Strategy: Engage students in role-playing to understand the scene.

Implementation:

- Assign students roles (bus driver, passengers, pedestrians, traffic lights).
- Have them act out **a busy road scene**, pausing to discuss what might cause an accident.
- Ask: **“How can we make this scene safer?”**

5. Story Walk

Strategy: Display key moments from the story around the classroom.

Implementation:

- Place images or short descriptions of:
 1. **Adam and Laila on the bus**
 2. **The accident scene**
 3. **Adam telling his father his idea**
- Have students walk from station to station, discussing what happens at each step.

6. Think-Pair-Share

Strategy: Encourage students to reflect and discuss in pairs.

Implementation:

- Pause at key moments in the story and ask: **“What do you think Adam will do next?”**
- Have students discuss their thoughts with a partner.
- Invite a few pairs to share their ideas with the class.

7. Interactive Storytelling

Strategy: Keep students engaged by involving them in the story.

Implementation:

- Ask students to **make sound effects** (bus engine, traffic sounds) during the story.
- Use **hand gestures or props** to show key elements like stop signs and crosswalks.
- Have students **vote** on what they think Adam should do next in the story.

Purpose of Reflection: Encourages critical thinking and helps students connect story themes to lesson objectives.

Reflection Answers

1. How did Adam feel after the bus stopped suddenly?

 **Correct Answer: (b) Scared**

2. Based on the story, what are the main concerns of Laila and Adam about the city?

 **Correct Answer: (b) The cars are too fast**

Strategies to Use with This Part

1. Think-Pair-Share

How to Implement:

- Ask students to think individually about the **reflection questions** and their answers.
- Pair them up to **discuss their reasoning** with a partner.
- Invite pairs to **share their answers** with the class, comparing different viewpoints.

2. Role-Playing Responses

How to Implement:

- Assign students different roles: **Adam, Laila, the bus driver, or the bus supervisor.**
- Have them act out a scene where Adam and Laila react to the bus stopping suddenly.
- Discuss their emotions and **why Adam might feel scared.**

3. Text Evidence Challenge

How to Implement:

- Ask students to **point to parts of the story** that describe Adam's emotions or concerns about the city.
- Have them **explain how the text supports their answer.**
- Discuss how **understanding a character's feelings helps us connect to the story.**

4. Open-Ended Questioning

How to Implement:

- Pose open-ended questions like:
 - **"How would you feel if you were in Adam's place?"**
 - **"What would you do to make your city safer?"**
- Encourage students to **share creative ideas about how technology could help prevent accidents.**
- Highlight the importance of **thinking about solutions like Adam did.**

Let's think: Safe City Picture Sorting Activity

How to Implement:

1. Prepare Materials:

- Gather pictures of **safe city features** (e.g., traffic lights, pedestrian crossings, organized roads) and **unsafe city elements** (e.g., streets with no traffic signals, congested intersections, reckless driving).
- Print or display these images on a screen for group activity.

2. Introduce the Idea:

- Explain in simple terms that cities need special rules and designs to keep people safe.
- Ask, "**What do you think makes a city safer for people and cars?**"

3. Sorting Activity:

- Give each student or small group a mix of pictures showing both **safe** and **unsafe** city conditions.
- Ask them to sort the images into two groups:
 - ✓ **Safe City Features**
 - ✗ **Unsafe City Features**
- As they sort, guide their thinking with simple questions like:
 - "**How do traffic lights help people cross safely?**"
 - "**What might happen if there were no traffic signs?**"

4. Class Discussion:

- Once students have sorted their pictures, ask them to share **one example** from each category.
- Discuss why certain features make a city **safer or more dangerous** for people.
- Relate the discussion back to Adam and Laila's concerns in the story.

5. Wrap-Up:

- Talk about **one or two ways** students can help keep their city safe (e.g., following pedestrian rules, using crosswalks, waiting for green lights).
 - Reinforce the idea that **small changes** help make a city a better place for everyone.
-

Engagement Tip:

- Use **stickers** or **thumbs-up voting** to let students decide on the best safe city features.
 - Allow students to **draw or design** their own "perfect safe city" as an extension activity.
-

Explore Activity: Learning About City Safety

Teacher Role:

- Introduce **Sparky**, the friendly bot, and explain that he will help students learn about **problems in cities and what causes accidents**.
- Guide students through the interactive elements (helping characters reach their destination and watching the video).
- Ask **open-ended questions** to encourage thinking:
 - **"What dangers do you see in a busy city?"**
 - **"How can we make sure everyone stays safe?"**
- Facilitate discussion about **real-life city problems** (e.g., too many cars, careless driving, lack of traffic signs).

Expected From Students:

- Interact with the **game or activity** that helps characters navigate the city safely.
 - Watch the **video** and describe key takeaways.
 - Identify the **main causes of accidents** in cities.
 - Suggest **solutions** for making their city safer.
-

Strategies to Use with This Part

1. Interactive Game Exploration

How to Implement:

- Guide students through the **game or digital activity** where they help characters reach their destination safely.
- Pause at key moments to ask:
 - **“What should the character do next to stay safe?”**
 - **“Why is this road more dangerous than the other?”**
- Encourage students to **verbalize their decisions** before making a move.

2. Watch & Discuss

How to Implement:

- Show the **video about city safety** and have students **describe what they noticed**.
- Ask simple guiding questions like:
 - **“What happened in the video that was unsafe?”**
 - **“How did people in the video solve city problems?”**
- Write down student ideas and compare them with Adam’s solutions from the story.

3. Problem-Solution Brainstorm

How to Implement:

- Divide the class into **small groups** and assign each a common city safety problem (e.g., speeding cars, missing crosswalks).
- Ask each group to **think of a solution** and share it with the class.
- Create a **class chart** of city safety problems and solutions.

4. Design a Safe City Map

How to Implement:

- Give students a **blank city map** and have them **add safety features** (e.g., stop signs, crosswalks, bike lanes).
- Encourage them to **color and label** their safe city design.
- Let them **present** their maps and explain their safety choices.

Engagement Tip:

- Use **stickers or stamps** as rewards when students suggest creative safety solutions.
 - If possible, invite a **guest speaker** (e.g., a local traffic officer) to discuss **real-world city safety measures**.
-

Assessment

Objective:

Students will demonstrate their understanding of **crossing the street safely and recognizing traffic signs** by answering multiple-choice questions and participating in a hands-on activity.

Answers to Assessment Questions:

1. **What should you look for when crossing the street?**
☒ **A) A zebra crossing**
2. **When is it important to hold hands with your daddy?**
☒ **B) When it's time to cross the road**
3. **What should you do before you cross the street?**
☒ **A) Look right and left**

Materials Needed:

- **Picture Cards** of different **traffic signs** (e.g., stop sign, pedestrian crossing sign, speed limit sign).
- **Toy Cars and Small Figurines** to act out street crossing.
- **A Large Poster or Flashcards** showing common traffic signs.
- **Floor Tape** to create a zebra crossing in the classroom.

Step-by-Step Instructions:

1. Introduction: Understanding Traffic Signs

- Show a **Stop Sign** and ask, *"What does this sign tell cars to do?"*
- Show a **Pedestrian Crossing Sign** and ask, *"What should people do when they see this sign?"*

2. Answering the Questions:

- Read the multiple-choice questions aloud.
- Have students point to the correct **picture card** that matches the answer.

Strategies to Use with This Part

1. Traffic Sign Sorting Activity

How to Implement:

- Give students **picture cards of different traffic signs**.
- Ask them to **sort** them into two groups:
 - 1-**Signs for Cars** (e.g., speed limit, stop sign).
 - 2-**Signs for People** (e.g., pedestrian crossing, school zone sign).
- Discuss why these signs are important.

2. Practice Crossing the Street

How to Implement:

- Use **floor tape** to create a pretend zebra crossing.
- One student plays a **car driver**, while others take turns crossing safely.
- Encourage students to **look right and left** before crossing.

3. Traffic Sign Hunt

How to Implement:

- Hide **traffic sign flashcards** around the classroom.
- Let students find them and say what each sign means.
- Ask, *"Where have you seen this sign before?"* to connect learning to real life.

Engagement Tip:

- Reward participation with **stickers or high-fives** to encourage learning in a fun way!

Make a poster challenge

Objective:

Students will create a **safety poster** to remind people how to be careful when **crossing the street** and **driving** safely.

Materials Needed:

- **Large poster paper** or drawing sheets.
 - **Crayons, markers, or colored pencils.**
 - **Pre-made cutouts** of traffic signs, crosswalks, and traffic lights.
 - **Glue and stickers** for decoration.
-

Step-by-Step Instructions:

1. Discussion: Why Do We Need Street Safety?

- Ask: *"Why do we need to be careful when crossing the road?"*
- Show a picture of a **busy street** and discuss **safe vs. unsafe** crossing behavior.

2. Brainstorming: What Should Be on the Poster?

- Write down students' ideas on the board. Some key messages could be:

- **Stop, Look, Listen**
- **Always use the crosswalk**
- **Drivers should watch for pedestrians**
- Let students **choose a message** for their poster.

3. Creating the Posters

- Hand out materials and let students **draw and color** their safety posters.
- Encourage them to **add pictures** of traffic signs, crosswalks, and people crossing safely.

4. Sharing and Displaying

- Let each student present their poster and explain their safety message.
- Hang the posters in the **classroom or hallway** as a reminder for everyone!

Strategies to Use with This Part

1. Group Poster Collaboration

How to Implement:

- Divide students into **small groups**.
- Each group works together to create **one big poster** with multiple safety tips.
- Encourage teamwork and **let them present together**.

2. Role-Playing Safe & Unsafe Crossing

How to Implement:

- Use **toy cars and pretend roads** to show **safe vs. unsafe** ways to cross the street.
- Let students **act out** being pedestrians and drivers.
- Ask: *“What should you do before crossing?”* and let students practice saying **“Stop, Look, Listen!”**

3. Classroom Safety Walk

How to Implement:

- Take students on a **short walk around the school** (if possible).
- Look for real **traffic signs, crosswalks, and safety features**.
- Ask: *“Where should we cross? What signs do we see?”* to connect learning to real life.

Engagement Tip:

- Let students **vote** for their favorite poster and give a **“Safety Star” award** to encourage participation!

UNIT 01

Lesson 02

Chapter 1: Design

Engineering design process

Objectives:

1-Understand the **Engineering Design Process** and its importance in solving problems.

2-Focus on the **Ask** and **Brainstorming** stages to identify user needs and generate ideas.

3-Work collaboratively to explore different solutions without fear of failure.

Challenge Questions of the Lesson

- 1-Why is it important to ask questions before designing something?
- 2-How can brainstorming help us create better ideas?
- 3-How can technology make cities safer for everyone?

SEL relation

- 1-Self-Awareness: Encourages students to recognize their own problem-solving strengths.**
- 2-Responsible Decision-Making: Helps students evaluate different solutions before choosing one.**
- 3-Collaboration: Promotes teamwork during brainstorming and idea-sharing activities.**

Engineering design Process:

Quickly review the phases of the Engineering Design Process, then encourage students to focus on the Ask and Brainstorming stages until they settle on a project idea. Remind them that there are no right or wrong answers for this step—it's all about exploring possibilities!

Engineering Design Process LESSON 01

Name:
Project:

Ask

Imagine

Improve

Plan

Create

Objectives:

- 1-Use a **ruler** to measure accurately.
 - 2-Identify and describe **basic shapes** in city structures.
 - 3-Differentiate between **2D and 3D shapes** in real-world applications.
 - 4- Design a simple city layout using **Tinkercad** and craft materials.
- ☐ Understand the importance of **traffic signs, roads, and urban planning** for a safe city.

Challenge Questions of the Lesson

- 1-Why do we need traffic signs and pedestrian crossings in a city?
- 2-How does using basic shapes help us design buildings and roads?
- 3-How can measuring accurately with a ruler improve our designs?
- 4-Why is planning important before building a city model?

SEL relation

- 1-**Collaboration:** Students work in groups to design and build a city together.
- 2-**Problem-Solving:** Encourages critical thinking about urban planning and traffic safety.
- 3-**Creativity & Confidence:** Designing and presenting their own city helps build self-expression and confidence.
- 4-**Responsibility:** Understanding how city planning affects safety teaches accountability in real-world applications.

1. Engage

Teacher Role:

- Introduce the lesson by discussing city design and traffic safety.
- Use visuals and questions to spark curiosity.
- Guide students in making predictions about city planning and traffic control.
- Encourage participation and different viewpoints.

Expected From Students:

- Actively engage in discussions and predictions.
- Observe and analyze visuals.
- Share ideas and respond to classmates.
- Connect the lesson to real-life experiences.

Strategies and Implementations

1. Preview with a Hook

- Show an image of a **crowded, unorganized city** versus a **well-planned city** with roads, traffic lights, and pedestrian crossings.
- Ask: "**Which city looks safer? Why?**"
- Briefly discuss how good design and traffic control improve safety.

2. Make Predictions

- Reveal the lesson title: "**Designing a Safer City.**"
- Ask: "**What do you think makes a city safe for people?**"
- Write down student ideas and revisit them later in the lesson.

3. Set the Scene with a Visual

- Show a city map with streets, traffic signals, and crosswalks.

- Play a short **sound clip of city traffic** to immerse students.
- Ask: "**What do you notice about how people and cars move in the city?**"

Let's think

Concept: Measurement and Shape Recognition

Students will explore the importance of accurate measurements in city planning and understand the difference between 2D and 3D shapes.

Objective:

- Understand why accurate measurement is important in design.
- Recognize the difference between 2D and 3D shapes.
- Apply these concepts to real-world city planning.

Materials Needed:

- Rulers
- Printed shape cutouts (squares and cubes)
- Grid paper for drawing measurements

Step-by-Step Instructions:

1. Engage (Building Curiosity)

Teacher Role:

- Show two simple buildings: one with correct proportions and one with incorrect measurements.
- Ask: "*What do you notice about these two buildings? Which one looks better and why?*"
- Relate this to city design—why careful measurement matters.

Student Actions:

- Observe and compare the buildings.
- Share thoughts on how measurement affects design.

2. Explore (Hands-On Activity)

Strategy: Measurement Challenge

- Give students rulers and ask them to measure objects in the classroom.
- Discuss: *"What happens if we don't use the ruler correctly?"*
- Let them experiment with drawing two versions of the same shape—one measured correctly, one measured incorrectly.

Shape Sorting:

- Show students a **square (2D)** and a **cube (3D)**.
- Ask: *"How are they different?"*
- Guide them to recognize depth and volume in 3D objects.

3. Explain (Connect to Real Life)

- Discuss how architects and engineers use rulers to design safe buildings and roads.
- Relate back to city planning—accurate measurement ensures **properly spaced roads, sidewalks, and traffic signs**.

4. Challenge Questions:

1. *What happens if we don't measure correctly when designing a city?*
 - Possible Answer: Buildings and roads might not fit together well.
 2. *How is a square different from a cube?*
 - Possible Answer: A square is flat (2D), and a cube has depth (3D).
-

Explore

Students will practice measuring objects in their environment and explore the difference between 2D and 3D shapes through interactive activities.

Objective:

- Develop measurement skills using real objects.
- Identify and compare different shapes.
- Apply spatial reasoning to real-world objects.

Materials Needed:

- Rulers
- Objects around the classroom (books, pencils, blocks, etc.)
- Tablets or computers (to scan QR codes for interactive activities)

Step-by-Step Instructions:

1. Engage (Building Curiosity)

Teacher Role:

- Ask: *“How do we know the size of something?”*
- Show a small and a large book and ask students how they could measure them.
- Encourage them to look around the classroom and find different objects to measure.

Student Actions:

- Observe and discuss ways to measure objects.
- Predict which objects are bigger/smaller before measuring.

2. Explore (Hands-On Activity)

Activity 1: Measuring Hunt

- Instruct students to pick **three objects** in the classroom.
- Using a ruler, they should **measure each object** and compare sizes.
- Discuss: *“Which object is the longest? The shortest? How do we know?”*

Activity 2: Interactive Shape Exploration

- Guide students to scan the **QR code** to explore **shapes in a digital game**.
- Let them classify different shapes as **2D or 3D**.
- Ask: *“What shapes do you see in the classroom that match the ones in the game?”*

3. Explain (Connect to Real Life)

- Discuss how measuring is used in city planning, such as for **building heights, road widths, and playground spaces**.
- Explain that **engineers and designers** use shapes to create objects like **traffic signs, bridges, and buildings**.

4. Challenge Questions:

1. *Why do we need to measure objects?*
 - Possible Answer: To know their size and use them correctly in design.
 2. *How can we tell if a shape is 2D or 3D?*
 - Possible Answer: 2D shapes are flat, and 3D shapes have depth.
-

Assessment: Measuring and Decomposing Shapes

Students will apply their measurement skills and explore how complex structures can be broken down into simple shapes for design.

Objective:

- Measure basic geometric shapes accurately.
- Decompose complex structures into simpler shapes.
- Apply design thinking using digital tools like Tinkercad.

Materials Needed:

- Rulers
- Grid paper or whiteboards
- Tablets or computers with Tinkercad access

Step-by-Step Instructions:

1. Focus (Measuring a Shape)

Teacher Role:

- Show the **triangle** and ask: *“How can we measure this shape?”*
- Guide students to measure the **base and height** of the triangle using a ruler.
- Discuss why measuring accurately is important in design.

Student Actions:

- Use a ruler to measure the triangle.
- Record their measurements and compare with classmates.

2. Practice (Breaking Down a Structure)

Activity: Decomposing a House into Shapes

- Display the **3D house model** and ask: *“What shapes can we see?”*
- Help students identify **rectangles (walls), triangles (roof), and cylinders (chimney)**.
- Guide them to **sketch** the house using these basic shapes.

Extension:

- If digital tools are available, students can **design their house in Tinkercad** using rectangles and triangles.

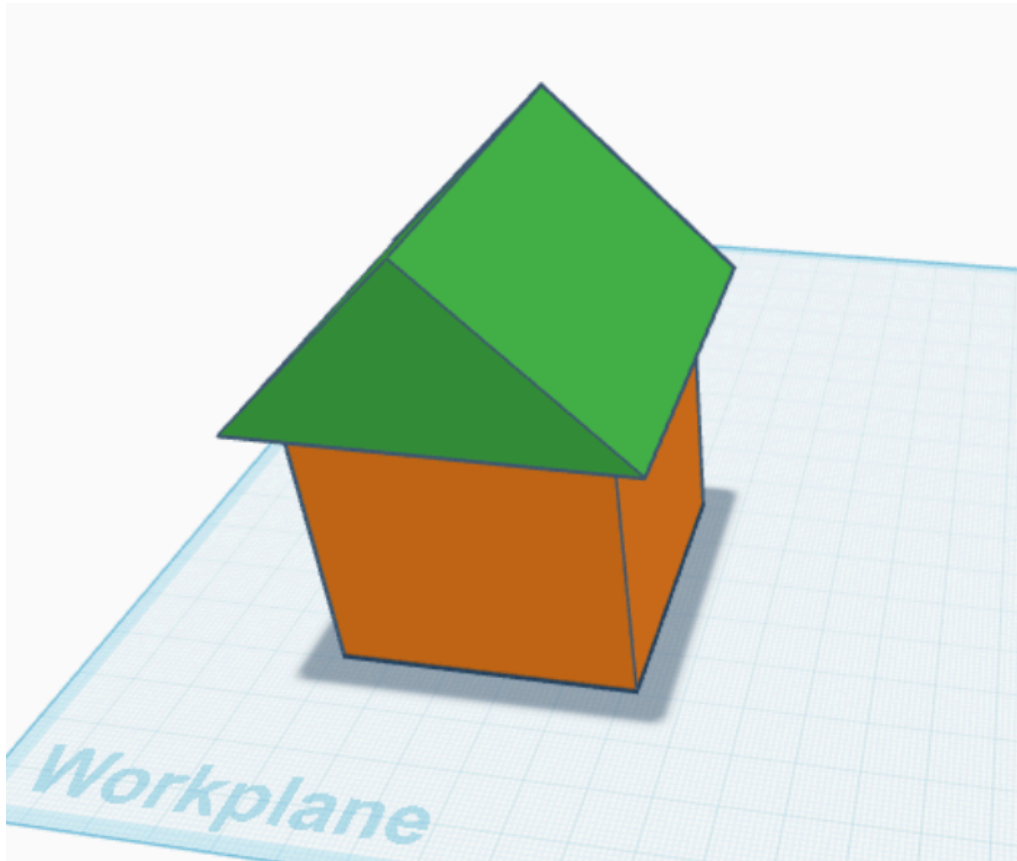
3. Challenge Questions:

1. *Why do we measure shapes before building something?*
 - o Possible Answer: To ensure accuracy and correct proportions in design.
2. *How can we turn a house into simple shapes?*
 - o Possible Answer: By identifying rectangles, triangles, and other basic forms.
 - o **3D design of house on Tinkercad**
Video link:
<https://www.youtube.com/watch?v=SpJCBGuJLJE>

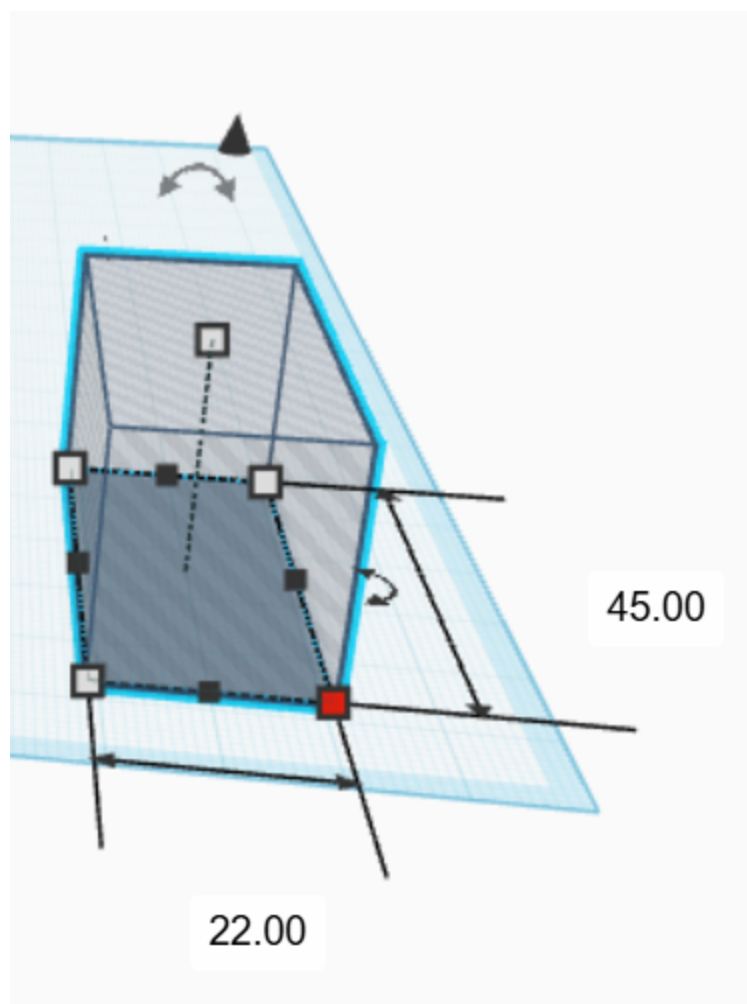
Steps:

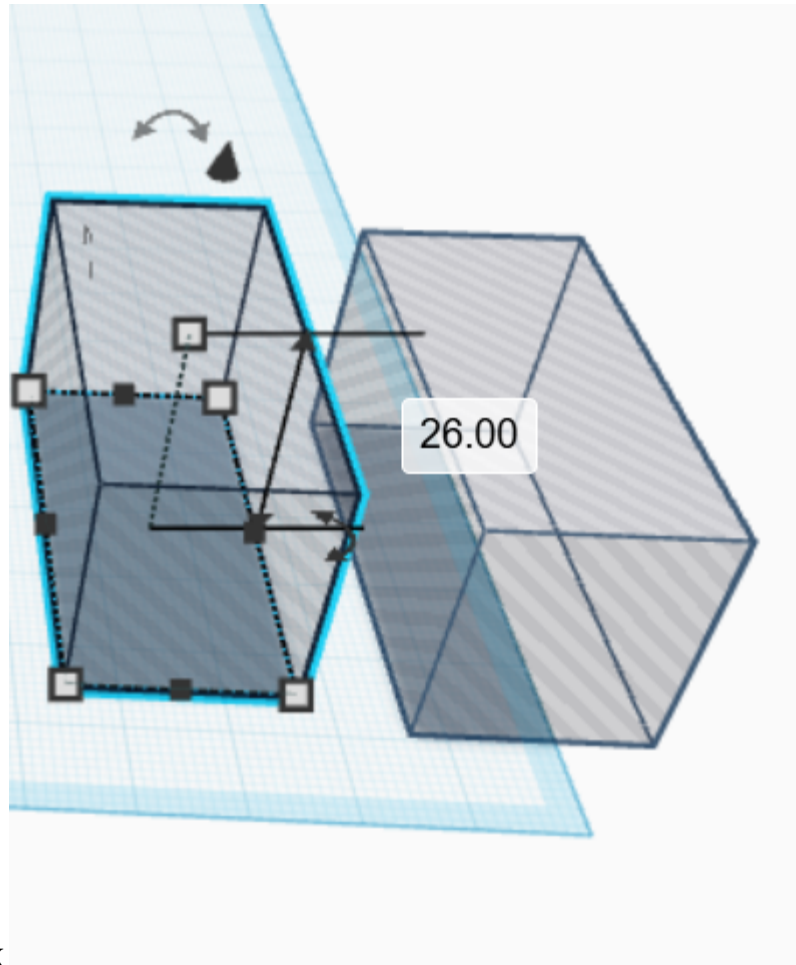
1-Create 50*50*50 orange box.

3-Align,group,then set color to be multicolor



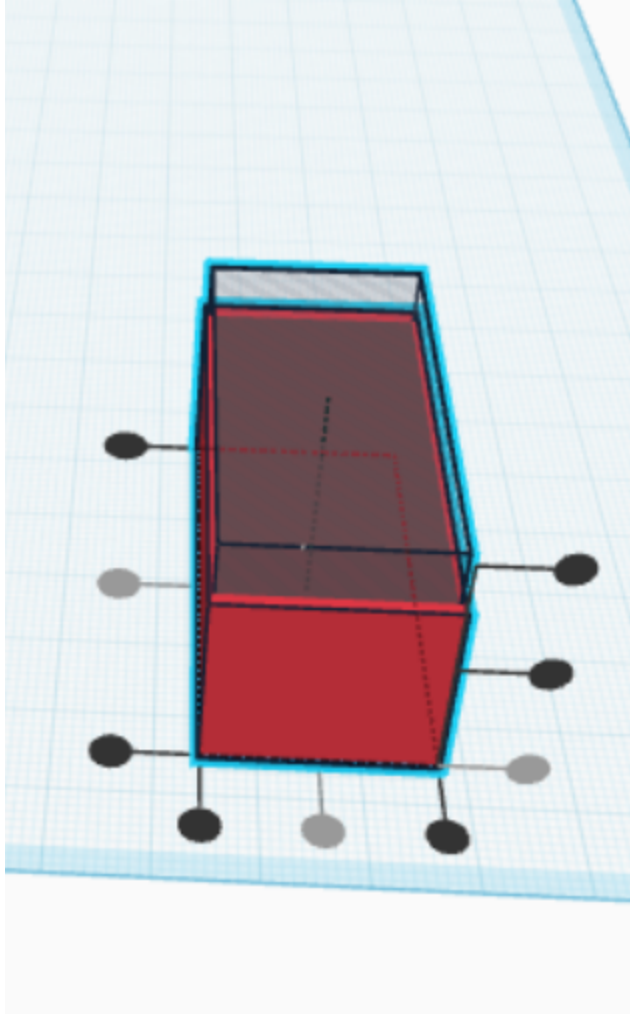
4-Create 48*23 red box



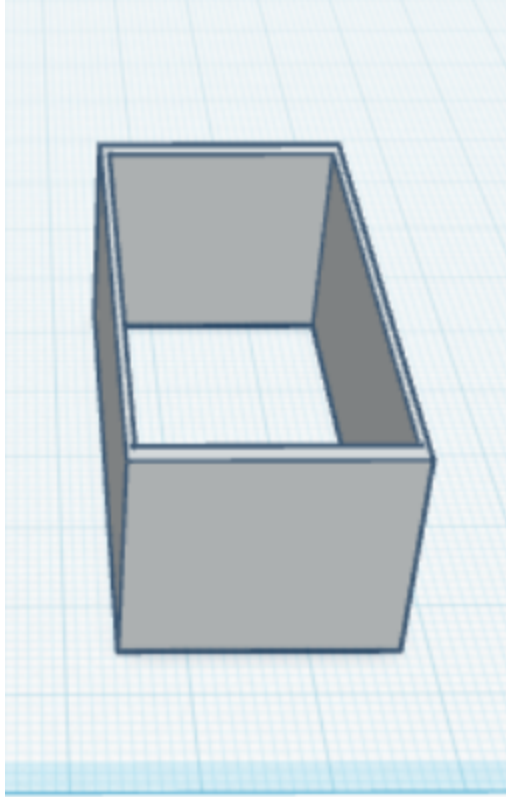


6-copy the hole box

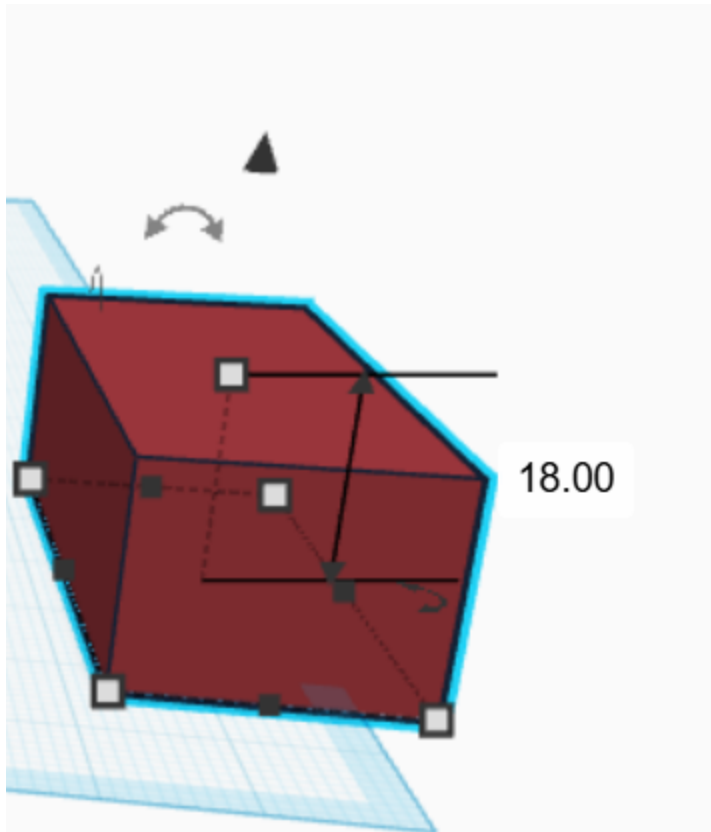
7-align red box and hole box together



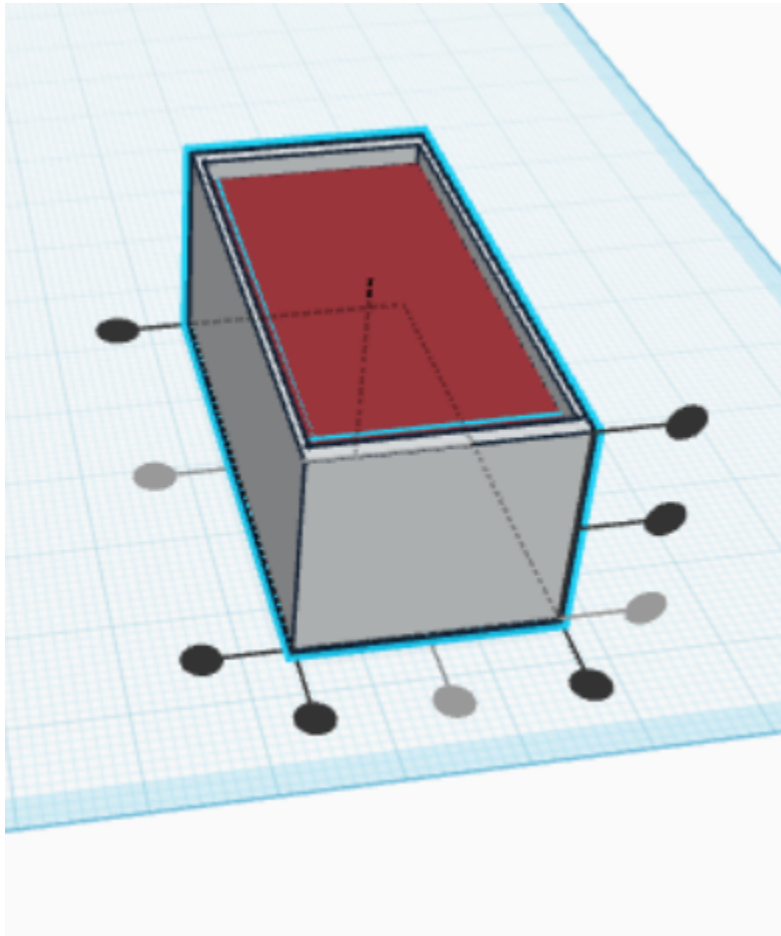
8- Group them to have this shape, and change color to grey.



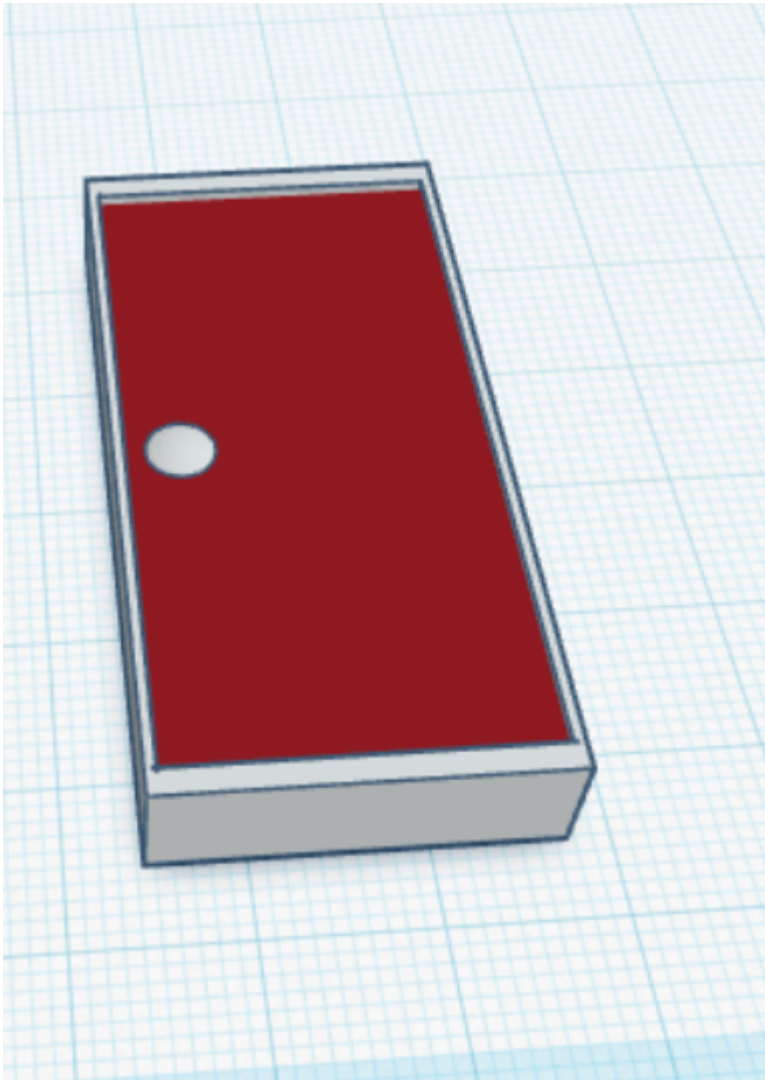
9-Make the other hole box solid and change color to dark red



10-Align 2 shapes and group them

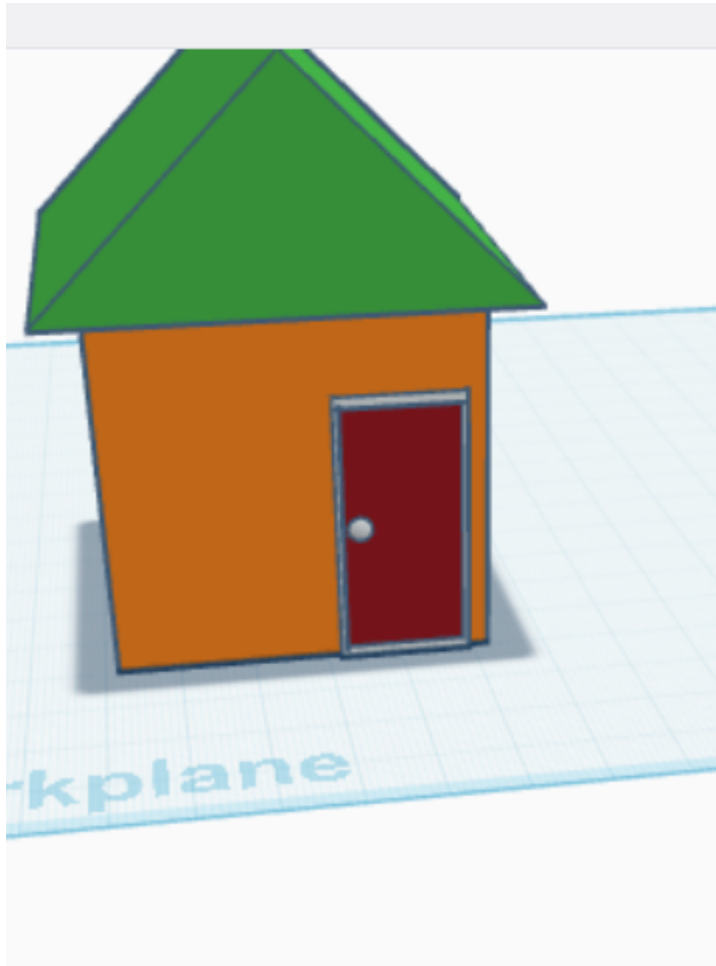


11-Change hight to 7 and add a sphere to represent door handle then

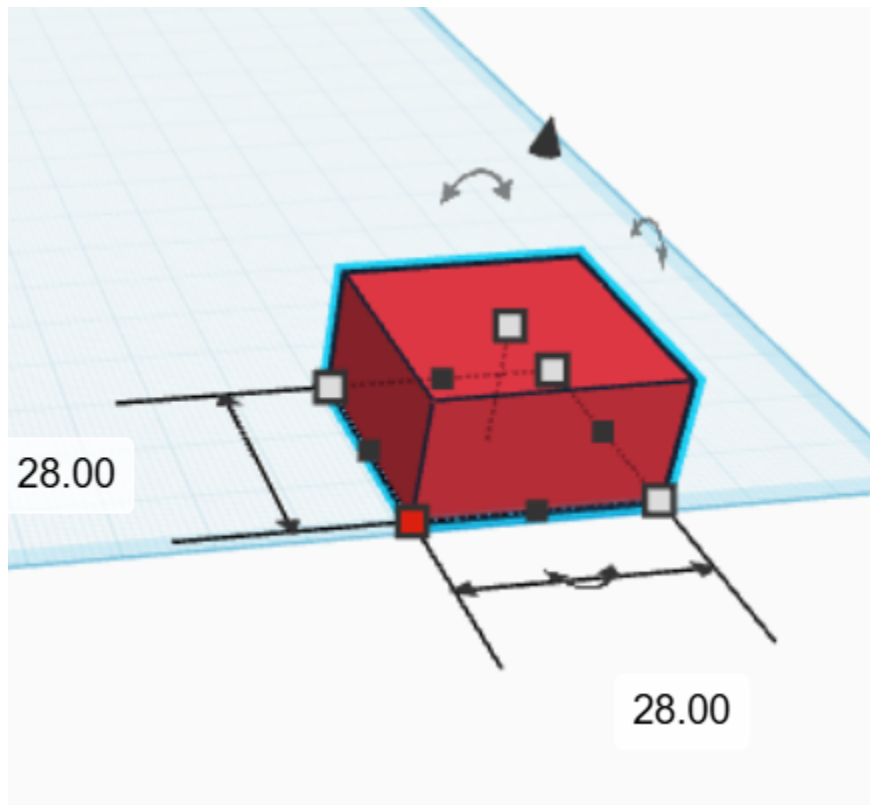


group

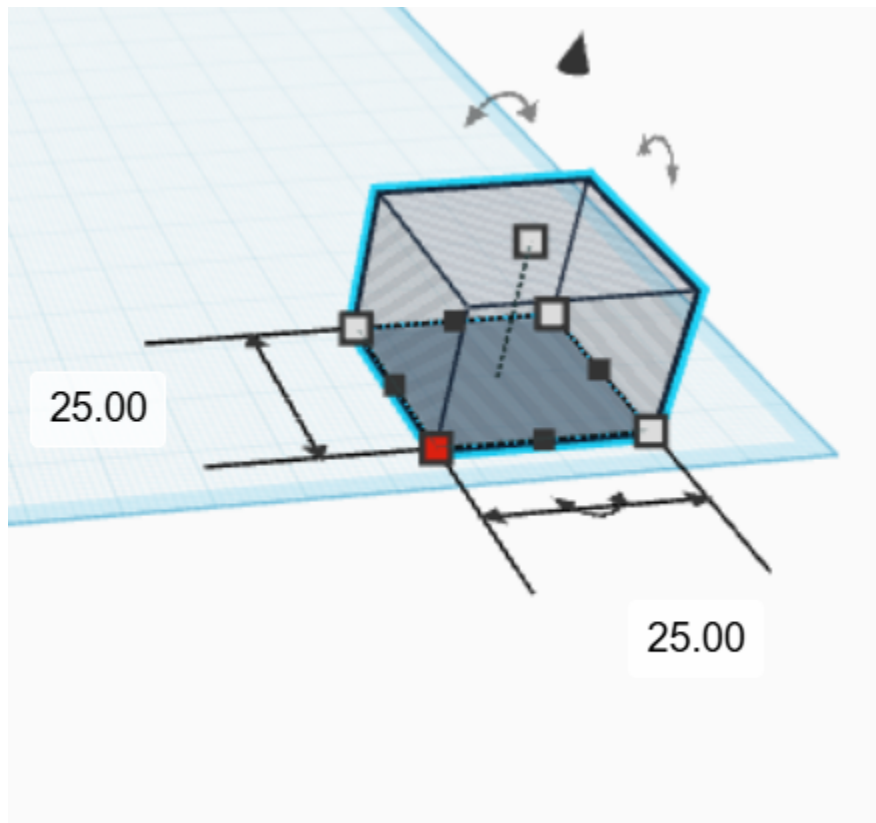
12-Rotate and adjust door to fit in the house

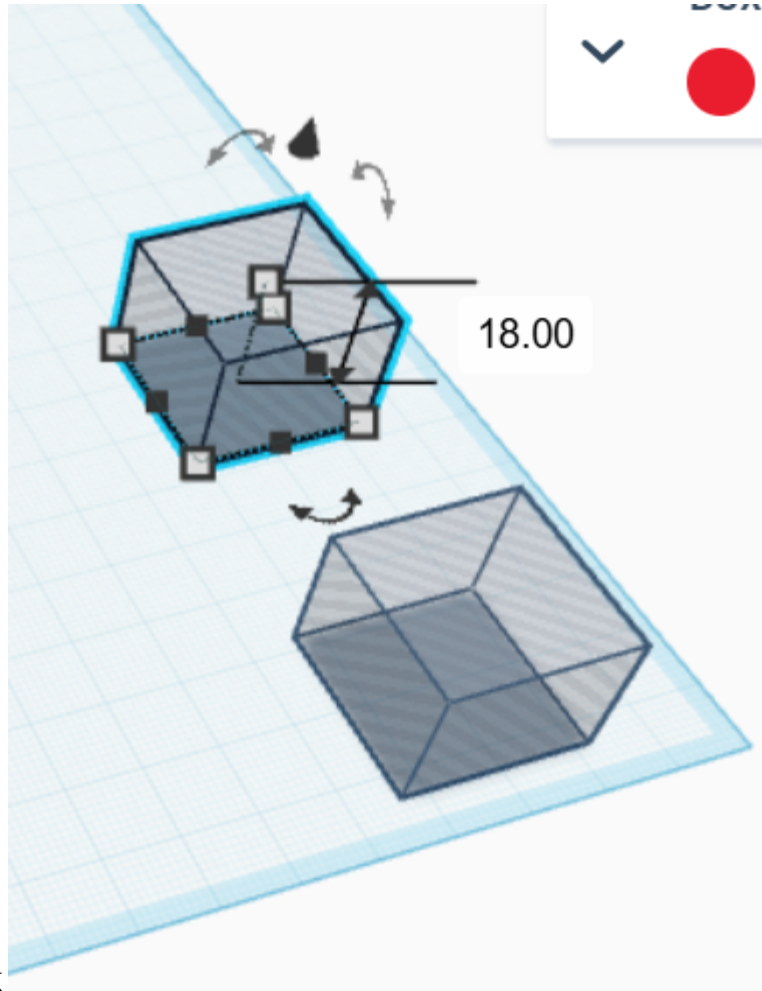


13-Creaet 28*28*15 red box



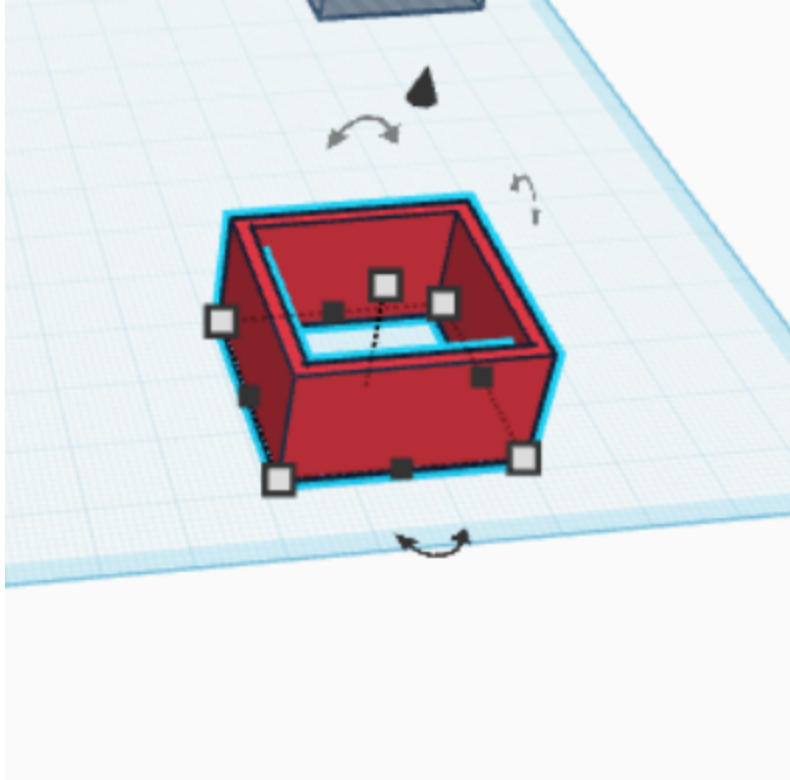
14-Create 25*25*18 hole box



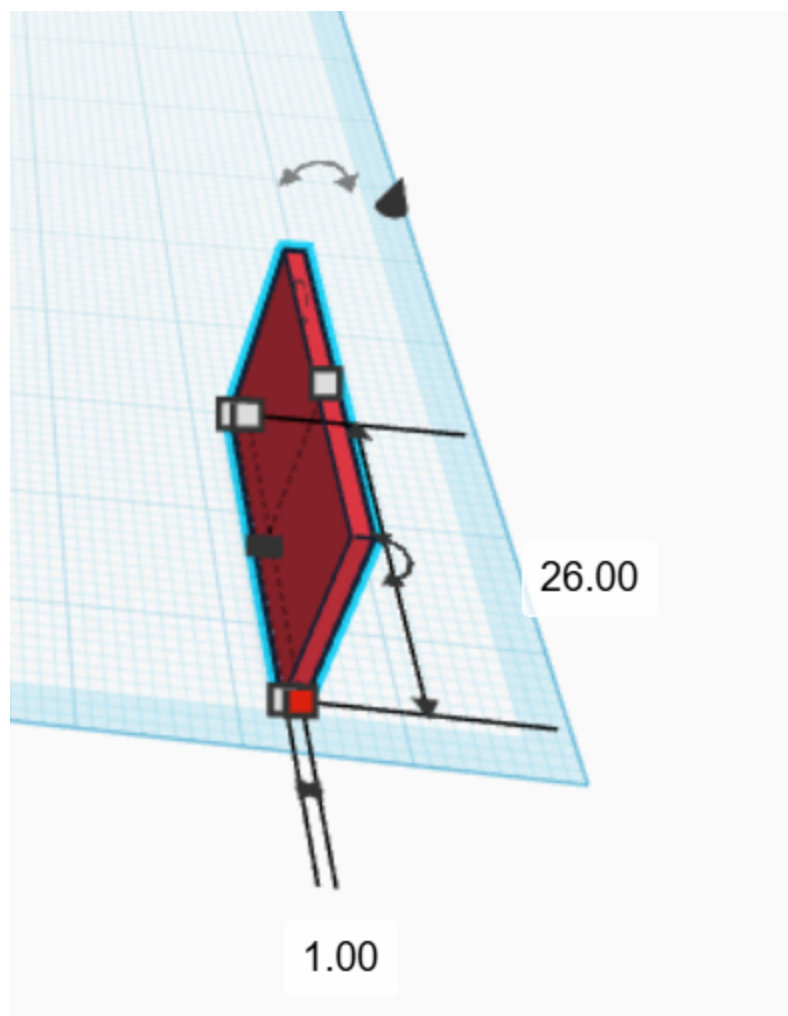


15-Copy the last box

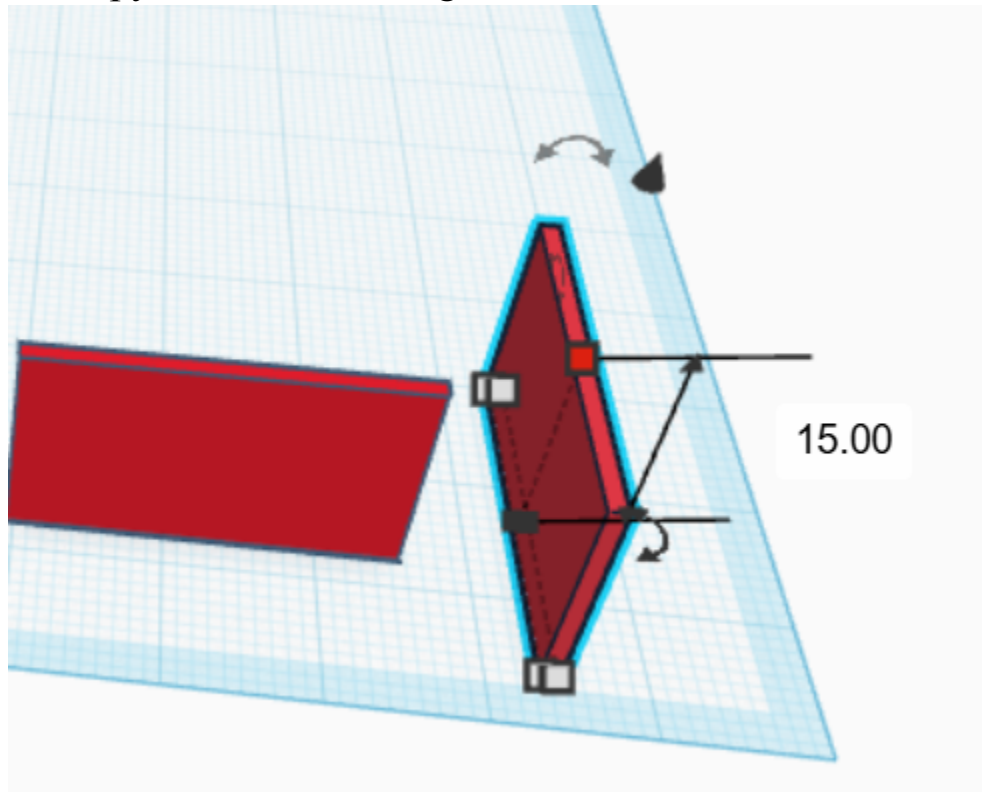
16-Align then group them together to get this shape



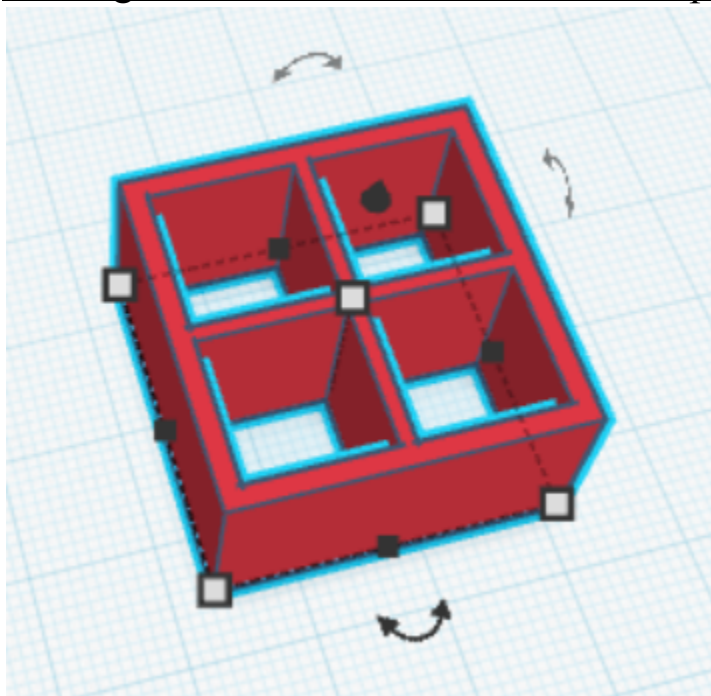
17-Create 26*1*15 red box



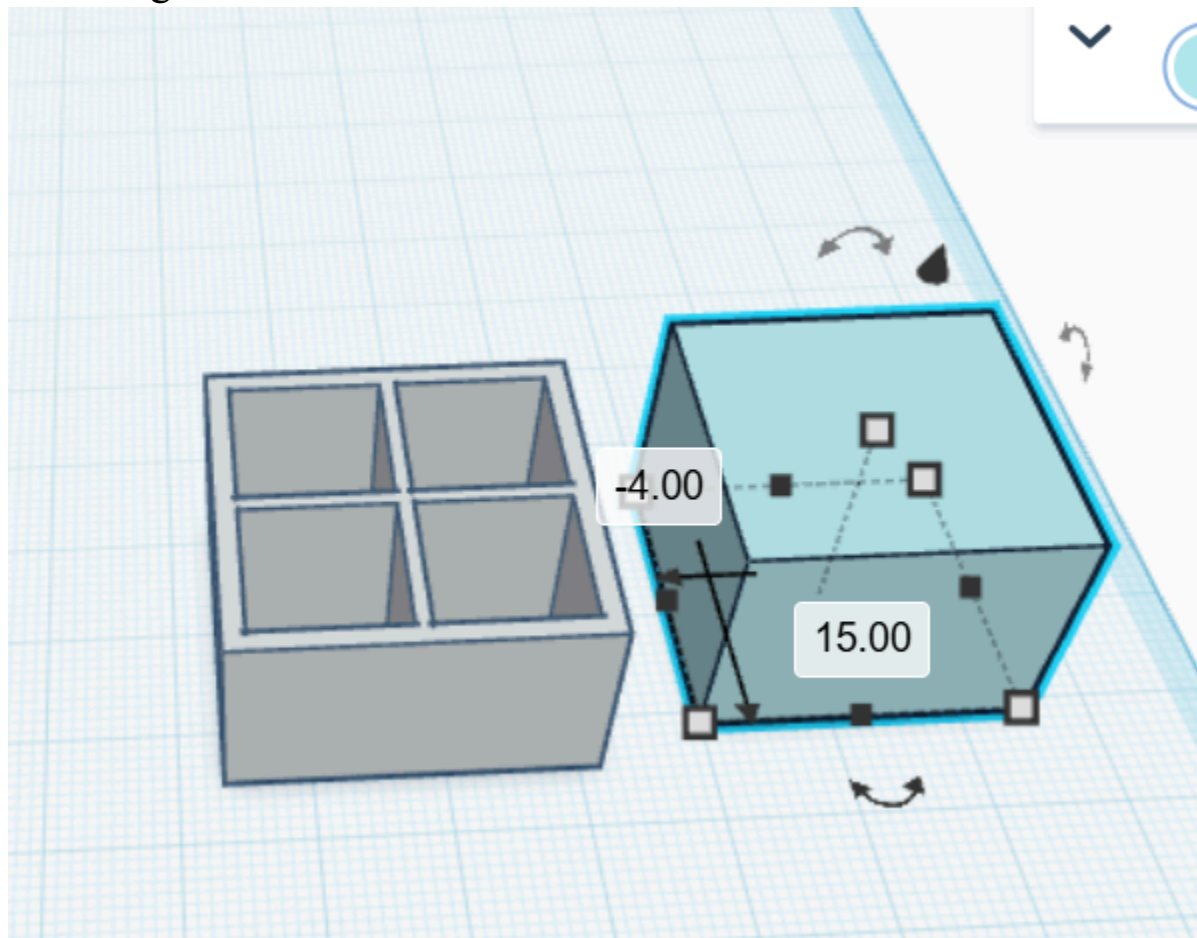
18-Copy and rotate 90 degree



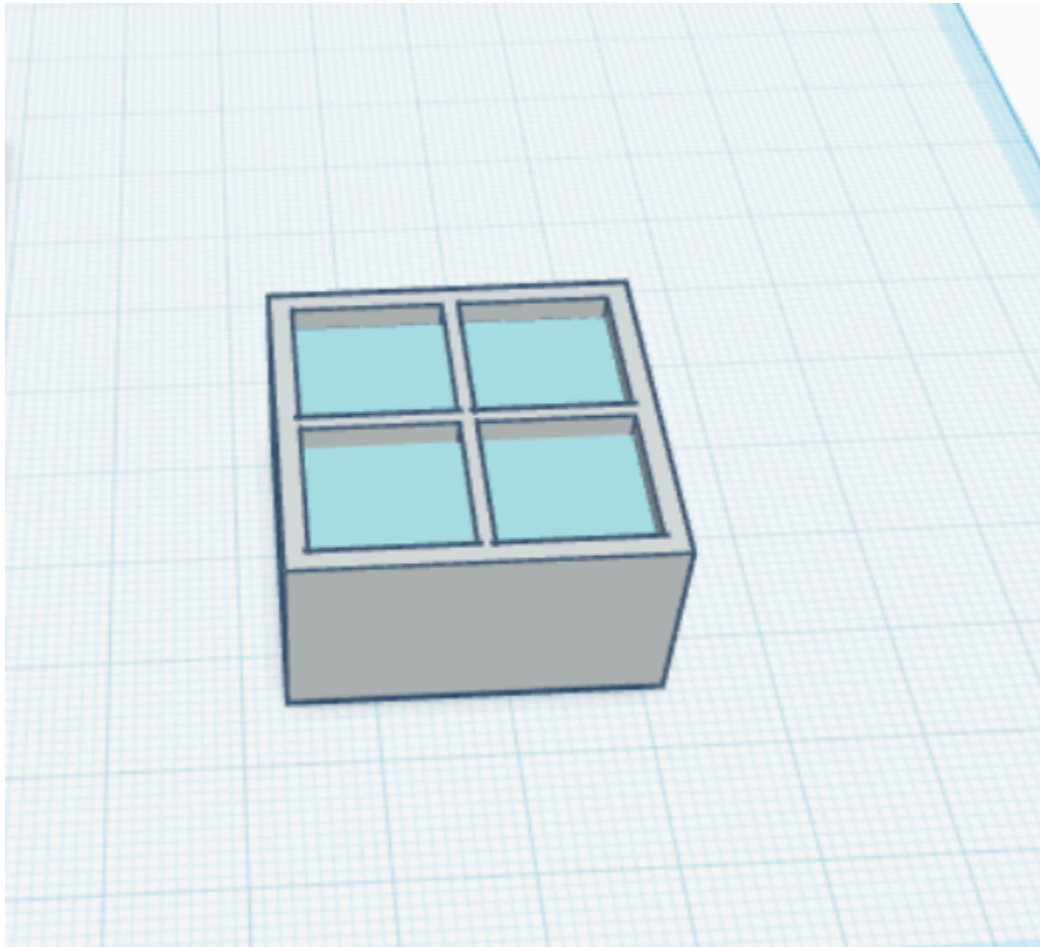
19-Align three of them to create this shape



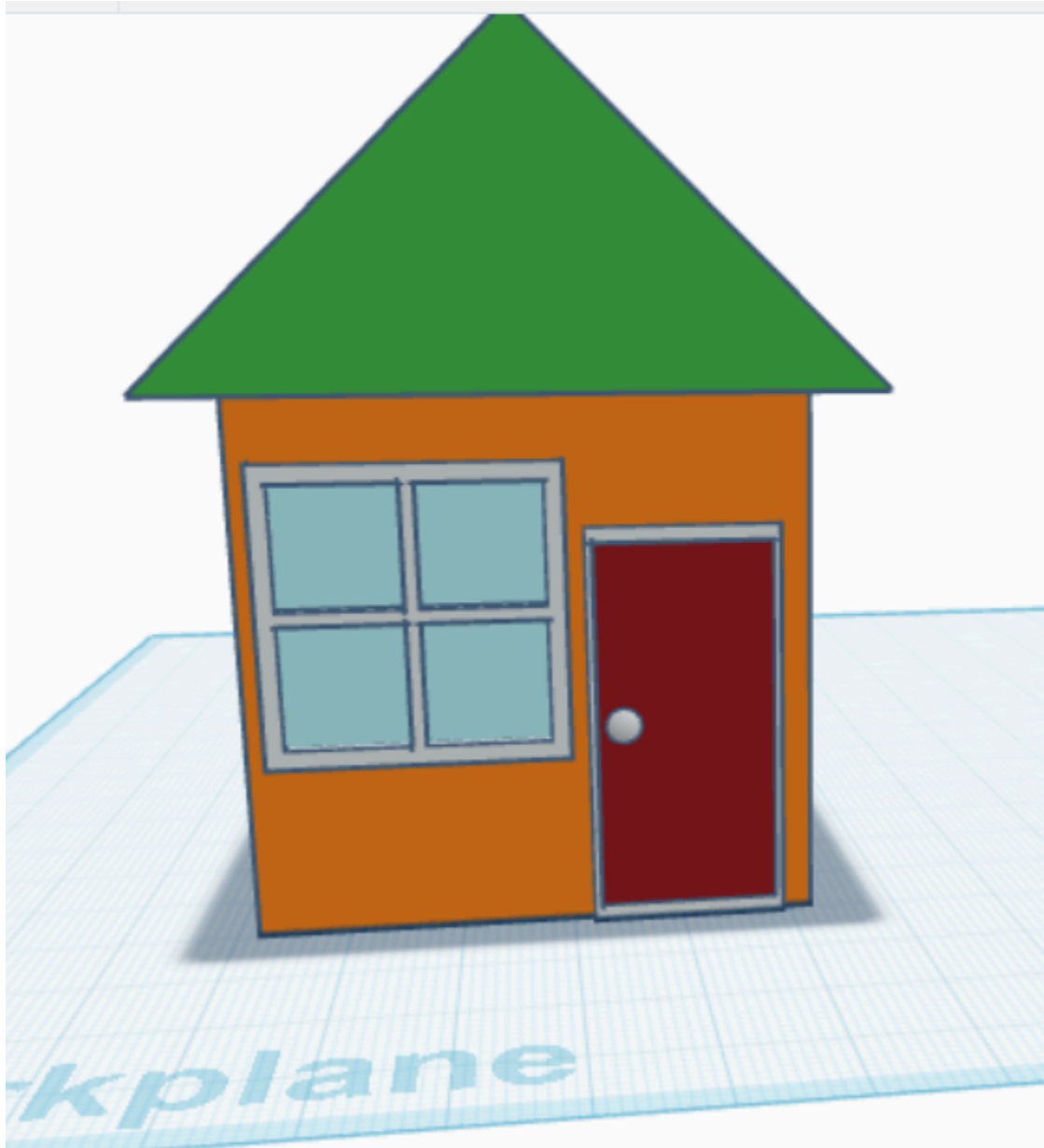
20-Change colors



21-Align, Adjust height, then group to look like this



22-Rotate, and adjust window to look like this



23-Copy to add another 2 windows on the side



Showcase: Creating a City Prototype

Students will bring their designs to life by constructing a city model with buildings, roads, and traffic signs while applying their knowledge of shapes and measurements.

Objective:

- Apply the **Engineering Design Process** to city planning.

- Construct a **city prototype** using basic shapes.
- Demonstrate understanding of **2D and 3D shapes** in real-world design.
- Use **Tinkercad** for basic digital design.

Materials Needed:

- Grid paper or foam boards
- Craft materials (cardboard, glue, scissors, markers)
- Rulers and measuring tools
- Tablets or computers with **Tinkercad**

Step-by-Step Instructions:

1. Connecting to the EDP Process

Teacher Role:

- Remind students to **return to their Engineering Design Process (EDP) plans** and refine their ideas before building.
- Ask: “*What features should our city have?*” (e.g., roads, traffic lights, pedestrian paths).
- Guide them in organizing their prototype using their design sketches.

2. Showcase: Building the City Prototype

Student Actions:

- Use **basic shapes** (rectangles for buildings, lines for roads, circles for roundabouts).
- **Assemble the city model** using craft materials.
- **Add details** like traffic signs, lights, and pedestrian crossings.
- **(Optional)** If digital tools are available, students can **recreate their city using Tinkercad**.

3. Reflection: “Now I Can...”

- Measure with a ruler.
- Identify and describe basic shapes.
- Understand the difference between **2D and 3D shapes**.
- Use **Tinkercad** for design concepts.

UNIT 01

Lesson 01

Chapter 2: Design domain

Cross-cutting roads

Objectives:

- 1-Explain how ramp steepness affects the speed of an object.
- 2-Apply the Engineering Design Process (EDP) to create and modify a ramp.
- 3-Identify real-world uses of ramps in different environments.
- 4-Recognize the importance of ramps in accessibility and transportation.

Challenge Questions of the Lesson

- 1-How does the steepness of a ramp affect how fast an object moves?
- 2-Why are ramps used instead of stairs in some places?
- 3-How can ramps be designed to help people with different needs?

1. Engage

Teacher Role:

- Introduce the concept of ramps by discussing where students have seen them in real life (e.g., playground slides, wheelchair ramps, or roads on a hill).
- Use an interactive approach by showing images of different types of ramps and asking, “What do you think these are used for?”
- Encourage students to share their thoughts on why ramps might be useful in a city.
- Guide students to think about ramps as **inclined planes**, a type of simple machine that helps make movement easier.
- Ask leading questions like:
 - “What happens when you roll a ball down a ramp?”
 - “How would a ramp help someone pushing a heavy object?”
- Relate the topic to their smart city project by discussing how ramps improve accessibility and efficiency.

Expected From Students:

- Actively participate in discussions by sharing their observations and prior knowledge of ramps.
- Answer open-ended questions and think critically about why ramps are useful.
- Make connections between ramps and real-world applications, such as in transportation, construction, and accessibility.
- Engage with visuals, storytelling elements, and class discussions to deepen their understanding.

Strategies to use with this part:

Preview with a Hook:

- **Strategy:** Start with a thought-provoking question or image.

- **Implementation:**

- Show an image of a city street with and without ramps.
- Ask: “How do you think ramps change the way people move in a city?”
- Facilitate a brief discussion on accessibility and efficiency.

Make Predictions:

- **Strategy:** Encourage students to predict the importance of ramps in their smart city model.

- **Implementation:**

- Ask: “How would a city without ramps be different from one with ramps?”
- Let students discuss their ideas with a partner and share their responses.

Set the Scene with a Visual:

- **Strategy:** Use images or a short video clip to illustrate ramps in action.

- **Implementation:**

- Display a short video of a ball rolling down different inclined surfaces.
- Ask: “What do you notice about how the ball moves? Does it slow down or speed up?”

Act It Out:

- **Strategy:** Have students physically engage with the concept of ramps.
 - **Implementation:**
 - Set up a small ramp in the classroom using books and a board.
 - Let students roll different objects down and observe the effects.
 - Encourage students to explain their observations.
-

Let's Think

Teacher Role:

- Guide students to analyze the two ramps in the images and predict how they will affect speed.
- Ask: *"How do you think the shape or steepness of a ramp affects how fast an object moves?"*
- Encourage students to think about real-world examples (e.g., playground slides, wheelchair ramps, skateboarding ramps).

- Facilitate discussion about **force and motion** before conducting hands-on activities.

Expected From Students:

- Observe and compare the two ramps in the images.
- Make predictions about which ramp would make an object move faster.
- Explain their reasoning based on prior knowledge or intuition.
- Write down their thoughts before testing their ideas.

Discussion Prompts:

- *Which ramp do you think will make the car go faster? Why?*
- *What forces do you think are acting on the car?*
- *What happens if we change the height or slope of a ramp?*

Think-Pair-Share:

- **Strategy:** Encourage students to share their ideas with a partner before discussing as a class.
 - **Implementation:**
 1. Give students time to write their thoughts.
 2. Have them discuss their ideas with a partner.
 3. Ask volunteers to share different predictions and explanations.
-

Explore – Ramp Experiment

Teacher Role:

- Guide students to build and test ramps with toy cars.

- Emphasize the relationship between ramp steepness and speed.
- Use open-ended questions to encourage observations.
- Facilitate comparisons between Ramp A and Ramp B in the images.
- Reinforce upward vs. downward motion and its effect on speed.
- Introduce the video resource to support learning.

Expected From Students:

- Build ramps and test how the steepness affects the car's speed.
- Observe and compare the results of different ramp angles.
- Record their findings on which ramp makes the car go faster.
- Connect their observations to forces like gravity and friction.
- Watch the provided video and reflect on key takeaways.

Guiding Questions:

1. What do you notice about the speed of the car on Ramp A vs. Ramp B?
2. How does the angle of the ramp affect the car's movement?
3. What happens when the car moves up the ramp instead of down?
4. How does this relate to real-world ramps (e.g., wheelchair ramps, skateboarding ramps, or roller coasters)?

Watch & Discuss:

- Play the video linked in the book.
 - Discuss how it relates to students' findings.
 - *Ask: Did the video confirm or challenge your predictions? Why?*
-

Step-by-Step Instructions for Explore Activity

Materials Needed:

- **Toy car** (or similar rolling object).
 - **Cardboard, books, or wooden blocks** (to create ramps).
 - **Measuring tape or ruler** (optional, to measure ramp height).
 - **Stopwatch or timer** (optional, to measure speed).
 - **Printed images of ramps A & B** (from the student book, if needed).
 - **Projector or device** (to display and watch the ramp video).
-

1. Preparation:

- Set up two ramps: **one steep (Ramp A)** and **one gentle (Ramp B)**.
 - Ensure students have materials to build and test ramps.
 - Set up the **YouTube video** link for later discussion.
-

2. Introduction:

- Explain that students will **explore how ramps affect speed**.
 - Ask: *What do you think will happen if we roll a car down a steep ramp vs. a gentle ramp?*
 - Show the **Ramp A & B images** in the book and discuss predictions.
-

3. "Do It" Activity:

Ramp Testing:

1. Have students **build or observe two ramps** (steep & gentle).
2. Roll the **toy car down each ramp** and observe which is faster.
3. (Optional) **Time the car's movement** for a measurable comparison.
4. Discuss: *Why do you think one ramp made the car move faster?*

Upward vs. Downward Motion:

1. Have students **push the car UP the ramps** and describe what happens.
 2. Discuss **force and effort** when moving objects uphill vs. downhill.
 3. Ask: *How does this relate to wheelchair ramps or skateboarding?*
-

4. Discovery Phase:

- Compare results: **Which ramp was faster? Why?**
 - Discuss **gravity and incline** in simple terms.
 - Encourage students to connect findings to **real-world ramps** (slides, roads, accessibility ramps).
-

5. Watch & Reflect:

- Play the **video** linked in the book.
- After watching, discuss:
 - *Did the video confirm your predictions? Why or why not?*

- *How do engineers use ramps in real life?*
 - *What happens if ramps are too steep or too flat?*
-

Assessment Guide

Objective:

- Reinforce students' understanding of **force, push, pull, and motion**.
 - Provide an opportunity for students to **apply their knowledge** through a multiple-choice quiz.
 - Use the **Einstein Bot** character to summarize key concepts before assessing.
-

Step-by-Step Instructions

1. Preparation:

- Ensure students have their **books open to the assessment page**.
 - If needed, provide **scrap paper or whiteboards** for students to jot down answers.
 - Optionally, prepare a **small demonstration** of push/pull using classroom objects (e.g., opening a door, pulling a backpack).
-

2. Review Key Concepts with Einstein Bot:

- Read aloud the **Einstein Bot summary** on the page.
- Emphasize:
 - *Push = Moving something away using force.*

- *Pull = Bringing something closer using force.*
 - *Motion = When something moves from one place to another.*
 - Use simple **real-world examples** (e.g., pulling a rope, pushing a toy car).
-

3. Conduct the Assessment:

- **Read each question aloud** (if needed).
 - Have students **circle their answers** in the book or **answer verbally**.
 - Allow time for **peer discussion** before revealing the correct answers.
-

4. Answer Key & Discussion:

1. What do we call a push or a pull? → (B) Force

- *Explain:* A force is needed to make things move, stop, or change direction.

2. Which of these is an example of a pull? → (B) Opening a drawer

- *Discuss:* Which actions require pulling vs. pushing? Demonstrate if possible.

3. What can happen to an object in motion? → (C) It can speed up, slow down, or change direction.

- *Relate to ramps:* When a car moves down a steep ramp, it speeds up. If a ramp is flat, it slows down.
-

5. Reflection & Wrap-Up:

- Ask: *Can you think of other real-life examples of push, pull, and motion?*
 - Optional: **Exit Ticket Activity** – Have students **act out** a push/pull motion or **draw a real-life example**.
-

Practice Guide

Step-by-Step Instructions

1. Preparation:

- Provide students with **paper, pencils, rulers**, and optional craft materials (cardboard, tape, glue) for ramp design.
 - Ensure students have access to their **EDP worksheet** to document their process.
-

2. Design Your Own Ramp (Hands-On Activity):

- **Introduce the Task:** Students will **design their own ramp** and imagine how it fits into a **city model**.
 - Encourage them to consider:
 - **Purpose:** Will it be for pedestrians, cars, or accessibility?
 - **Incline:** Steep or gentle? How does that affect speed?
 - **Materials:** What would it be made of in real life?
 - Students **sketch their ramp** and label key features.
-

3. Connect to the EDP (Engineering Design Process):

- **Ask:** What step of the EDP are we at now? (Design phase)

- Guide students to **add details** to their EDP worksheet:
 - **Problem:** Why is the ramp needed?
 - **Plan:** Sketch the design.
 - **Create:** If materials allow, they can build a **small model**.
 - **Test & Improve:** Ask, *How could you change the design to improve it?*
-

4. Real-World Application:

- Discuss where students have seen ramps **in real life**:
 - **Home:** Driveways, wheelchair ramps.
 - **School:** Stairs with ramps, playground slides.
 - **Community:** Skate parks, parking garages.
-

5. Reflection & “Now I Can...” Check-In

- Read the **three learning goals** at the bottom of the page.
- Have students **check off** the statements they feel confident about.
- Ask reflection questions:
 - *How did designing your ramp help you understand ramps better?*
 - *How does the steepness of a ramp change how things move?*
 - *Where else can we use ramps to help people?*

Objectives:

- 1-**Understand** the role of speed bumps in road safety and city planning.
- 2-**Analyze** how traffic control measures impact drivers and pedestrians.
- 3-**Apply** design principles to create a **speed bump model** using Tinkercad or physical materials.
- 4-**Evaluate** the effectiveness of different speed bump placements in an intersection.
- 5-**Document** their design choices in the **Engineering Design Process (EDP) journal**.

Challenge Questions of the Lesson

- 1-**How does the height of a speed bump affect cars and pedestrians?**
- 2-**Where should speed bumps be placed in a city to be most effective?**
- 3-**How can we balance road safety with driver convenience?**

SEL relation

Self-Awareness: Students reflect on their own experiences with road safety and how traffic control affects their daily lives.

Social Awareness: Understanding how road safety impacts all members of a community, including pedestrians, drivers, and cyclists.

Responsible Decision-Making: Encouraging students to make thoughtful design choices when placing speed bumps in their city model.

Collaboration & Communication: Sharing and discussing different viewpoints on urban safety and accessibility.

Engage (Stepping Stone)

Teacher Role:

Facilitate a discussion by introducing the concept of **electricity** and how it powers cities.

Ask students what they **already know** about electricity.

Use visuals or a short video showing **how cities light up at night**.

Encourage students to **predict** how they might add lights to their smart city models.

Expected From Students:

Participate in discussions about electricity and circuits.

Share prior knowledge or personal experiences with lights or electrical devices.

Make predictions about how they might integrate lights into their city models.

Preview with a Hook:

Strategy: Use an engaging question and real-life scenario.

Implementation:

- Show a **picture of a city at night** with lights and another city in darkness.
 - Ask: "**What do you notice? What would happen if our cities had no electricity?**"
 - Discuss the importance of **electricity in urban life**.
-

Make Predictions:

Strategy: Introduce the storyline of the lesson.

Implementation:

- Read aloud **Adam's dialogue** from the student book.
 - Ask: "**How do you think Adam can add lights to his city?**"
 - Jot down student ideas and revisit them later.
-

Set the Scene with a Visual:

Strategy: Use engaging media.

Implementation:

- Show a **smart city model** without lights vs. one with LED lights.
 - Ask: "**What differences do you notice? How do lights improve a city?**"
 - Play a **short sound clip** of a bustling, well-lit city.
-

Let's think

Teacher Role:

- Guide students to observe the **three objects (TV, table, washing machine)**.
 - Ask: "**Which of these objects need electricity to work?**"
 - Encourage students to **explain their reasoning**.
 - Ask: "**What do you think electricity is? Can you draw it?**"
 - _____
-

Student Tasks:

- **Circle the correct objects** (TV and washing machine).

- **Discuss why the table does not need electricity.**
 - **Write or draw their ideas** about electricity.
-

Discussion Prompts:

- **“Where does electricity come from?”**
 - **“What happens when there’s no electricity at home?”**
 - **“Can you think of more devices that use electricity?”**
-

Explore

1. Teacher Role

Guidance During the Activity:

- Introduce the lesson by discussing the importance of electricity in daily life. Ask students to list devices they use that require electricity.
- Explain the objective: understanding electricity and how to create a simple circuit on a smart pad.
- Guide students through observation-based questions:
 - "What do you notice about devices that need electricity?"
 - "Why do you think some objects need electricity while others don't?"

Clarification and Feedback:

- Ensure students understand the definition of electricity in a simplified way.
- Provide real-life examples of how electricity powers homes, schools, and cities.

- Encourage students to reflect on what life would be like without electricity.

2. Expected From Students

Active Participation:

- Engage in class discussions about electricity and its uses.
- Complete activities like identifying electrical vs. non-electrical devices.
- Observe and discuss how adding lights to their city model enhances functionality.

Collaboration:

- Work in small groups to identify electrical devices.
- Share insights on the role of electricity in their city models.

Critical Thinking and Communication:

- Explain why certain objects need electricity and others do not.
- Discuss possible energy sources for their smart city model.

Reflection and Application:

- Think about how electricity impacts their daily lives.
- Consider renewable energy sources for future cities.

3. Activity Overview

Materials Needed:

- LED lights
- Batteries
- Wires

- Electrical tape
- Instructional guide for circuit connection
- QR code linking to an educational video

Step-by-Step Activity Instructions:

1. Preparation:

- Distribute the instructional guide to students, explaining the components of a simple circuit.
- Ensure each group has the required materials to build a circuit.

2. Introduction:

- Introduce Lab Bot as their guide for the activity.
- Explain the purpose of the task: creating a simple circuit to power an LED light.
- Demonstrate how circuits work using the guide as a reference.

3. "Do It" Activity:

- Instruct students to follow the guide to connect the LED light to the battery using wires.
- Allow students to experiment with different connections while ensuring safety.
- Encourage students to scan the QR code and watch the video for further understanding.

4. Discovery Phase:

- Observe students' attempts and provide real-time feedback.
- Discuss why a closed circuit is necessary for the LED to light up.

- Ask students to troubleshoot if their LED does not turn on, prompting critical thinking.

5. Reflection:

- Have students explain in their own words how electricity flows in a circuit.
- Ask students how they could apply this knowledge in real-world scenarios, such as powering small city models.

4. Inquiry-Based Learning & Assessment

Encouraging Observations & Questions:

- Foster curiosity by asking, "What happens if we disconnect one part of the circuit?"
- Encourage students to explore how electricity powers different objects.

Collaboration:

- Promote teamwork by having students build and troubleshoot circuits together.
- Allow groups to present their findings and share their experiences.

Formative Assessment:

- Assess student participation and engagement.
- Review completed circuits to ensure understanding.
- Use student reflections to gauge comprehension of electrical concepts.

Assessment Exercise

Focus Activity – Matching Electrical Components

- Introduce the activity by explaining that students will match components to their correct names.
- Explain the importance of identifying key circuit components like LEDs, batteries, and wires.
- Guide students to analyze the images and select the correct answers.

Practice – Building a Simple Circuit on the Smart Pad

- Encourage students to apply their knowledge by creating a virtual circuit on the smart pad.
- Provide step-by-step guidance on how to connect circuit components correctly.
- Allow students to experiment and troubleshoot their digital circuit designs.

Assessment & Reflection

- Observe students' accuracy in matching components and completing the smart pad circuit.
 - Ask students:
 - "How does electricity flow in this circuit?"
 - "What happens if we remove a component?"
 - Use their responses to reinforce key electricity concepts.
-

Challenge Activity

Paper Circuit Flower:

- Encourage students to apply their learning by creating a paper circuit flower, combining creativity with electrical concepts.
- Provide students with paper, copper tape, LED lights, and coin cell batteries to design their flowers.
- Guide them in designing their circuit layout before assembling the flower.
- Explain the importance of a closed circuit for the LED to light up.
- Provide QR code access to a video tutorial for step-by-step guidance.
- Allow students to personalize their flowers with colors and decorations while ensuring the circuit remains functional.

Reflection:

- Discuss how electricity makes everyday life easier by enabling lighting and technology.
- Ask students how they can use their knowledge to design something useful for their smart city.
- Encourage students to brainstorm other creative ways circuits can be used in everyday life.

Learning Outcomes:

- Identify electricity and its uses.
 - Recognize simple electronic components.
 - Construct a basic circuit on a smart pad.
 - Apply electrical concepts to a creative project.
-

Objectives:

- 1-**Understand** the role of speed bumps in road safety and city planning.
- 2-**Analyze** how traffic control measures impact drivers and pedestrians.
- 3-**Apply** design principles to create a **speed bump model** using Tinkercad or physical materials.
- 4-**Evaluate** the effectiveness of different speed bump placements in an intersection.
- 5-**Document** their design choices in the **Engineering Design Process (EDP) journal**.

Challenge Questions of the Lesson

- 1-**How does the height of a speed bump affect cars and pedestrians?**
- 2-**Where should speed bumps be placed in a city to be most effective?**
- 3-**How can we balance road safety with driver convenience?**

SEL relation

Self-Awareness: Students reflect on their own experiences with road safety and how traffic control affects their daily lives.

Social Awareness: Understanding how road safety impacts all members of a community, including pedestrians, drivers, and cyclists.

Responsible Decision-Making: Encouraging students to make thoughtful design choices when placing speed bumps in their city model.

Collaboration & Communication: Sharing and discussing different viewpoints on urban safety and accessibility.

Engage

Teacher Role:

- Introduce the topic by discussing **road intersections** and why safety measures are needed.
- Show visuals of **unsafe vs. well-planned intersections** to spark curiosity.
- Facilitate discussion by **asking students about their own experiences** with road safety.

Expected From Students:

- Actively participate in discussions about **road safety and traffic control**.
- Share personal observations about **intersections, crosswalks, and speed bumps**.
- Engage with images and **think critically** about how to improve road designs.

Preview with a Hook

Strategy: Start with an engaging image and a thought-provoking question.

Implementation:

- Display an image of an **unsafe intersection with no signs or speed bumps** vs. a **well-designed one with clear markings and speed control**.
- Ask: **“Which intersection is safer? Why?”**
- Discuss briefly to connect students’ ideas with the lesson.

Make Predictions

Strategy: Introduce the problem through the character **Adam's observation**.

Implementation:

- Read Adam's statement: **"I heard a lot of accidents happen at crossroads. Is there a way we can make them safer?"**
- Ask students: **"What solutions can we think of to make intersections safer?"**
- Encourage them to jot down ideas that will be revisited during the design phase.

Set the Scene with a Visual

Strategy: Use a mix of pictures and real-world references.

Implementation:

- Show a picture of a **road without traffic control** and compare it to a **well-planned intersection with speed bumps and signs**.
- Ask students: **"What do you notice? What makes one safer than the other?"**
- Let students discuss and share observations.

Explore

Teacher Role:

- Guide students in designing their **intersecting road** and placing a **clams post** for speed regulation.

- Encourage hands-on learning by allowing students to **experiment with different road layouts** in their city models.
- Facilitate discussion on **why speed bumps are placed in certain locations**.
- Use the provided **video resource** to reinforce understanding.

Expected From Students:

- Actively participate in **designing their road intersection**.
- Discuss and decide where to place the **clams post** for optimal safety.
- Watch the video and reflect on **how speed bumps improve road safety**.
- Add their designed elements to the **planning section** of their project.

Hands-on Exploration

Activity: Design Your Intersection

Objective:

Introduce students to the **crossroad safety challenge** by applying **problem-solving skills and teamwork** to design and implement an electronic warning system.

Implementation:

1. **Step 1:** Students create a model of an **intersecting road** using materials like paper, cardboard, or digital tools.
2. **Step 2:** They identify **key spots for speed bumps or traffic signs**.

3. **Step 3:** Discuss **why these locations were chosen** and how they improve safety.

Materials Needed:

- **Buzzer** (to alert cars at the intersection)
- **LEDs** (to signal drivers on other roads)
- **Wires & Connectors**
- **Battery Pack or Power Source**
- **Breadboard** (for circuit assembly)
- **Switches or Sensors** (to detect vehicle movement)
- **Cardboard or Small Model Roads** (for prototyping)

Activity:

1. Introducing the Challenge:

- Explain that the solution involves adding a **buzzer** at the intersection and an **LED light** on top of each road.
- The buzzer will activate when a car passes, and the LED will signal drivers on other roads to **reduce accidents at crossroads**.

2. Exploring Materials (5 min):

- Provide students with the **kit and materials** for the project.
- Allow them time to **explore and identify** the components before starting the challenge.

3. Team-Based Design Task:

- Divide students into **three teams** (3–4 members per team).

- Each team will **design a crossroad** using the provided materials.
- **Scoring System:**
 - Best crossroad design: **100 points**
 - Second-best design: **75 points**
 - Third-best design: **50 points**

4. **Circuit Building & Competition:**

- Students will learn **how to connect an electric circuit** and the **difference between series and parallel circuits**.
- Each team must **correctly wire the circuit** to make the system functional.
- **Scoring System:**
 - First team to complete the circuit correctly: **100 points**
 - Second team: **75 points**
 - Third team: **50 points**

Watch & Reflect

Strategy: Use a **video resource** to reinforce the importance of speed bumps.

Implementation:

- Show the **QR code or video link** provided in the book.
 - After watching, ask:
 - **“How do speed bumps help control traffic?”**
 - **“Where should we place them in our city model?”**
 - Allow students to modify their designs based on what they learned.
-

Connecting to the Planning Phase

After designing their **intersections and speed bump placement**, students should:

- 1–**Update their Engineering Design Process (EDP) planning sheet.**
- 2–**Discuss improvements before finalizing their design.**

Assessment

Teacher Role:

- Facilitate a **class discussion** on the purpose and function of **speed bumps** using the multiple-choice questions as a guide.
- Ensure students understand the **importance of speed bumps** in urban planning.
- Guide students in the **hands-on design challenge** to create a **speed bump model** in Tinkercad or using physical materials.

- Encourage students to document their designs in their **Engineering Design Process (EDP) journal**.

Expected From Students:

- Answer the **multiple-choice questions** and discuss the correct answers.
 - Engage in the **speed bump design activity**, considering height, size, and placement.
 - Use **Tinkercad or other design tools** to create a **3D model** of a speed bump.
 - Reflect on how speed bumps contribute to road safety and urban design.
-

Assessment Breakdown

1-Focus: Concept Check

- **Activity:** Multiple-choice quiz to assess students' understanding of speed bumps.
- **Implementation:**
 - Read each question aloud and discuss the correct answer as a class.
 - Encourage students to **justify their answers** with reasoning.
 - Connect answers to **real-life examples** in their city or school.
 -

Multiple-Choice Questions & Answers

1-What do speed bumps do?

- ☐ A) Make cars go faster
- ☒ B) Help cars slow down
- ☐ C) Paint the road
- ☐ D) Clean the cars

2-What shape are speed bumps like?

- ☐ A) Squares
- ☐ B) Circles
- ☒ C) Small hills
- ☐ D) Big holes

3-Why are speed bumps important?

- ☐ A) To make roads pretty
- ☒ B) To keep people safe
- ☐ C) To make noise
- ☐ D) To sell more cars

2-Practice: Hands-on Design

- **Activity:** Design a speed bump for the city model.
- **Implementation:**
 - Guide students in **creating a digital or physical speed bump.**
 - Encourage them to **consider height and placement** for safety and effectiveness.
 - Ask students:

- **“What happens if the speed bump is too high or too low?”**
 - **“Where should we place it for maximum safety?”**
-

Final Step: Document & Reflect

Students should:

- **Add their speed bump design** to their EDP planning paper.
 - Reflect on **how speed bumps help make a city safer.**
 - Share their designs and **explain their placement choices.**
-

Now I Can...

Understand how a speed bump works.

Design a speed bump and add it to my city.

UNIT 02

Chapter 2: Electricity domain

Lesson 01

Function your speed bump

Objectives:

- 1-Differentiate between open and closed circuits and how they affect electricity flow.
- 2-Build and test a simple circuit using a buzzer, push button, and wires.
- 3-Apply the Engineering Design Process (EDP) to troubleshoot and improve their circuit.

Challenge Questions of the Lesson

- 1-What is the difference between an open and closed circuit?
- 2-How does the buzzer work, and why does it stop when the button is released?
- 3-How can circuits be used in real-world safety systems like speed bumps?

SEL relation

Self-Management: Encourages patience when troubleshooting circuits.

Collaboration: Strengthens teamwork and communication in group work.

Problem-Solving: Develops critical thinking by testing and improving designs.

Resilience: Builds perseverance through hands-on challenges.

Real-World Awareness: Connects circuits to safety applications in daily life.

1. Engage

Teacher Role:

- Introduce the lesson by connecting it to real-life scenarios involving road safety and alarms.
- Use visuals or a brief discussion about traffic signals, speed bumps, and warning systems to capture students' interest.
- Ask open-ended questions such as: *“What happens when a car drives over a speed bump?”* or *“How do traffic lights help drivers and pedestrians?”*
- Guide students to think about how technology, such as buzzers and buttons, can be used to improve safety.
- Encourage collaborative discussions where students can share experiences and predictions.

Expected From Students:

- Actively participate in discussions by sharing observations and prior knowledge about road safety and alarms.
 - Make predictions about how circuits can be used to create an alarm system.
 - Engage with visual prompts or simple demonstrations to connect real-world applications with the lesson.
 - Show curiosity by asking questions and exploring different possible solutions.
-

Strategies to Use with this part:

Preview with a Hook:

Strategy: Begin with a real-world example.

Implementation:

- Show an image or video of a busy intersection with and without warning signals.
- Ask: *“How do drivers know when to stop or slow down?”*
- Discuss the importance of alerts in preventing accidents.

Make Predictions:

Strategy: Connect students' prior knowledge to the new concept.

Implementation:

- Present the lesson's key question: *“How can we design a simple alarm system to warn drivers?”*
- Have students brainstorm possible ways alarms are used in daily life.

Set the Scene with a Visual:

Strategy: Use images or short clips to illustrate the concept.

Implementation:

- Display a diagram of a simple circuit with a buzzer.
- Ask: *“What do you think will happen when the circuit is completed?”*

- Encourage students to discuss the purpose of each component.

Think-Pair-Share:

Strategy: Facilitate peer discussions to deepen understanding.

Implementation:

- Ask students: *"Why do you think an alarm system is useful at a crossroad?"*
 - Have them share their ideas with a partner.
 - Invite volunteers to present their thoughts to the class.
-

2. Let's think & Explore

Teacher Role:

- Present the two circuit diagrams (A & B) and ask students to predict whether the lamp will light up in each case.
- Guide students to analyze the differences between an open circuit and a closed circuit.
- Encourage critical thinking by asking: *"What do you observe in each circuit? What makes the bulb light up?"*
- Facilitate a discussion about how electricity flows in a circuit and the role of switches in controlling the flow.
- Introduce real-life applications of open and closed circuits, such as light switches at home.

Expected From Students:

- Observe and analyze the two circuit diagrams.
- Predict and justify whether the lamp will light up in each case.

- Discuss their reasoning with peers, using prior knowledge of electric circuits.
 - Relate the concept of open and closed circuits to everyday experiences.
-

Strategies to Use with this part:

Think-Pair-Share:

Strategy: Encourage students to analyze and discuss their observations.

Implementation:

- Have students individually decide whether the lamp will light in each diagram.
- Pair them up to compare answers and justify their reasoning.
- Invite volunteers to explain their conclusions to the class.

Hands-On Demonstration:

Strategy: Reinforce the concept with a physical demonstration.

Implementation:

- Use a simple circuit kit with a bulb, battery, and switch.
- Demonstrate what happens when the circuit is open versus closed.
- Ask students to explain the observed outcome.

Watch & Reflect:

Strategy: Use the provided video to enhance understanding.

Implementation:

- Direct students to scan the QR code or watch the video as a class.

- Pause at key moments to discuss how the concepts relate to the circuits in the lesson.
 - Ask students to summarize what they learned from the video.
-

3. Assessment

Teacher Role:

- Reinforce key concepts by reviewing **electricity, open circuits, and closed circuits** with students.
- Guide students through the **Focus** section by asking them to recall prior knowledge and apply it to multiple-choice questions.
- Facilitate hands-on practice with the **Practice** section, where students create a simple circuit using a buzzer, push button, and wire.
- Observe students' responses and provide feedback to clarify misconceptions.
- Encourage students to **explain their reasoning** for each answer to strengthen their critical thinking skills.

Expected From Students:

- Demonstrate understanding of electricity and circuits by answering assessment questions.
- Apply knowledge to identify key components in a circuit and their functions.
- Engage in hands-on practice by building a simple circuit on the smart pad.
- Reflect on the purpose of speed bumps and their real-world applications.

Focus (Multiple-Choice Questions)

1. What provides energy in a circuit?

A) Wire

B) Switch

☒ C) Battery (Correct Answer)

D) Light bulb

Explanation: The battery serves as the power source in an electric circuit, providing the energy needed for current to flow.

2. What do speed bumps do?

A) Make cars go faster

☒ B) Help cars slow down (Correct Answer)

C) Paint the road

D) Clean the cars

Explanation: Speed bumps are designed to slow down vehicles and improve road safety by reducing speed in specific areas.

Practice (Hands-On Activity)

Activity: Building a Simple Circuit

- **Objective:** Students will construct a basic circuit using a **buzzer**, **push button**, and **wire** on the smart pad.
- **Steps:**
 1. Provide students with a smart pad circuit simulator or physical components.

2. Instruct them to **connect the battery, buzzer, and push button** to form a closed circuit.
 3. Have them **predict** what will happen when the push button is pressed.
 4. Allow students to **test their circuit** and observe the buzzer activating.
 5. Discuss **what happens if the circuit is broken** (open circuit) and how to fix it.
-

Strategies to Use:

Quick Concept Review:

Strategy: Reinforce key vocabulary before the assessment.

Implementation:

- Write the terms **electricity, open circuit, closed circuit** on the board.
- Ask students to define each term in their own words.
- Relate each concept to real-life examples (e.g., light switches at home).

Think-Pair-Share for Focus Questions:

Strategy: Allow students to discuss before selecting an answer.

Implementation:

- Students first **think** about their answers independently.
- They then **pair** with a classmate to compare responses and reasoning.

- Volunteers **share** their answers with the class, explaining their thought process.

Hands-On Practice for Circuit Building:

Strategy: Apply learning through hands-on exploration.

Implementation:

- Provide students with a smart pad simulation of a circuit.
- Guide them step by step to **connect the buzzer, push button, and wire** correctly.
- Ask students to **predict the outcome** before turning on the circuit.
- Discuss why the circuit works and what would happen if any component were missing.

Exit Ticket Reflection:

Strategy: Assess understanding through a final reflection.

Implementation:

- Ask students to complete a quick exit ticket:
"What did you learn about circuits today? What questions do you still have?"
 - Review responses to identify areas for further clarification.
-

4. Showcase

Teacher Role:

- Guide students in applying their knowledge by **enhancing the speed bump with an alarm system**.
- Encourage students to document their ideas and solutions in the **EDP (Engineering Design Process) paper**.

- Facilitate a discussion about the **importance of alarms in speed bumps** and their real-world applications.
- Check for understanding by reviewing the "**Now I Can...**" section with students.

Expected From Students:

- Design and describe how to **integrate an alarm system into the speed bump project**.
 - Use the **EDP paper** to document the process of designing, creating, and improving their circuit.
 - Demonstrate an understanding of **simple circuits, buzzers, push buttons, and circuit flow**.
 - Self-assess their learning by completing the "**Now I Can...**" checklist.
-

Activity: Adding an Alarm to the Speed Bump

Objective:

Students will modify their speed bump design by incorporating an alarm system, reinforcing their understanding of **electrical circuits and real-world problem-solving**.

Steps:

1. Brainstorm:

- Discuss how an alarm system can improve road safety.
- Identify the components needed for the alarm circuit (buzzer, battery, switch, wires).

2. Plan & Sketch:

- Have students sketch a simple diagram of their circuit.
- Guide them to **label each component** and explain its function.

3. Document in EDP Paper:

- Students will add this step to their **Engineering Design Process (EDP) documentation**.
- Encourage them to **describe challenges and possible improvements**.

4. Present & Reflect:

- Students can **share their designs** with peers.
- Facilitate a discussion: *How does the alarm help in real-world applications? What improvements can be made?*

Now I Can... (Self-Assessment Checklist)

- Create a simple electric circuit with a buzzer and push button.
- Understand the difference between open and closed circuits.

Teacher Reflection & Next Steps:

- Review student work and **assess understanding** of circuit components and their functions.
- Provide feedback on EDP documentation and **encourage iteration** on designs.
- Relate this activity to future lessons, such as **expanding circuits with sensors or automation**.

Objectives:

- 1-Identify a traffic light and recognize its basic visual components (red, yellow, green lights).
- 2-Understand the basic function of each color on a traffic light (stop, be ready, go/move).
- 3-Design a basic visual representation of a traffic light.
- 4-Integrate their traffic light design into the context of a previously created city.
- 5-Begin to understand that technology can be used to improve safety in communities.

Challenge Questions of the Lesson

- 1-Why do traffic lights have red, yellow, and green?
- 2-Where is one important place to put a traffic light in our city?
- 3-What's one *super* idea to make a traffic light even better?

SEL relation

Self-Management: Practicing following instructions during the color-by-number activity and the DIY traffic light creation. They also manage their materials and time during these tasks.

Social Awareness: Considering the needs of others in the community by thinking about how traffic lights make roads safer for everyone (cars and pedestrians).

Relationship Skills: Collaborating during the "Think-Pair-Share" activity and while creating their traffic lights. They learn to share ideas and work together.

Responsible Decision-Making: Understanding that traffic lights are a system of rules designed to help people make safe choices on the road. They consider the consequences of not following traffic signals.

1. Engage:

Teacher Role:

- Introduce the idea of making our pretend city roads safer by talking about real-life safety on roads in a way that connects to what young children experience (walking to school, being in a car).
- Use simple questions and visuals to get them thinking about what helps keep them safe near roads.
- Encourage them to guess what a "smart" traffic light might do in a fun and imaginative way.
- Create a friendly and open space where all children feel happy to share their ideas.
- Help guide their thinking with easy-to-understand explanations and gentle prompts.

Expected From Students:

- Share their own experiences of being near roads and what makes them feel safe.
- Look at pictures and listen to the story about making the city safer.
- Offer their own simple ideas about what a "smart" traffic light might be.
- Listen carefully to their friends' ideas.
- Try their best to think about how to make roads safer for everyone.

Strategies for implementation:

- **Preview with a Hook:**

- **Strategy:** Ask a question about their direct experience.
- **Implementation:**
 - Ask, "When you walk near a road, what helps keep you safe?" (Possible answers: holding hands, looking both ways, grown-ups, maybe even traffic lights they've seen).

- **Make Predictions:**

- **Strategy:** Share what Mrs. Sara and Laila say in simple terms.
- **Implementation:**
 - Say, "Mrs. Sara said, 'Our city is built! Now, how can we make the roads super safe for everyone?'"
 - Ask, "What are some things we could add to our roads to make them safer?" (Let them give simple answers).
 - Say, "Then, Laila had a great idea! She said, 'Maybe we can add a *smart* traffic light!'"
 - Ask, "Hmm, a *smart* traffic light! What do you think a *smart* traffic light might do?" (Encourage imaginative guesses - "Maybe it knows when cars are coming!", "Maybe it changes colors really fast!").

- **Set the Scene with a Visual:**

- **Strategy:** Use the picture of their pretend city.

- **Implementation:**
 - Point to the roads in the city picture. Ask, "Where do you think cars and people will be moving in our city?" (Focus on roads and intersections).
 - Ask, "What could happen if lots of cars and people are trying to go through the same place at the same time?" (Guide them towards the idea of things getting messy or unsafe).
 - **Think-Pair-Share (Simplified):**
 - **Strategy:** Simple partner talk.
 - **Implementation:**
 - Ask, "Think about waiting to cross the street with a grown-up. What helps you know when it's okay to walk?"
 - Tell them to talk to their friend next to them about their answer.
 - Ask a few children to share what they talked about.
-

Let's Think - Exploring the Visual Representation of a Traffic Light

2. Explore :

Teacher Role:

- Introduce the color-by-number activity as a fun way to discover what a traffic light looks like.
- Provide clear instructions on how to use the number key and coloring materials.
- Encourage careful coloring and accurate number connections.
- Facilitate a discussion after the activity about the colors and order of lights on a real traffic light.

Expected From Students:

- Carefully color the sections of the picture according to the number key (which the teacher will provide with corresponding colors).
- Connect the numbered circles in the key in sequential order (1 to 14).
- Observe the resulting image of the traffic light.
- Participate in a discussion about the colors and their order on a real traffic light.

Materials Needed:

- Student books with the "Let's Think" color-by-number page.
- Crayons, colored pencils, or markers in the colors corresponding to the numbers in the key (Teacher will need to pre-determine and communicate this key).
- Pencils for connecting the numbers.

Step-by-Step Instructions:

1. Preparation (Before the Activity):

- **Crucially, the teacher needs to decide on the color key for numbers 1 through 14.** For the traffic light itself, ensure that

the colors assigned to the three lights (top, middle, bottom) will correspond to the standard order (Red, Yellow, Green). The remaining numbers can be assigned other appropriate colors for the truck and background.

- Clearly write or display the color key for students (e.g., 1 = Red, 2 = Blue, 3 = Yellow, 4 = Green, etc.). Use simple color names that Grade 1 students can easily understand.
- Ensure each student has access to the necessary coloring materials and a pencil.

2. Introduction (Engaging with the Activity):

- Say something like: "Look at this fun picture! It has lots of numbers. We're going to be color detectives and number connectors to see a very important tool that helps keep our roads safe!"
- Explain that they need to color each numbered section with the color that matches that number in the color key.
- Show them the color key clearly.
- Explain that after they finish coloring, they need to take their pencil and connect the numbered circles on the right side of the page, starting from 1 and going all the way to 14.

3. "Do It" Activity (Coloring and Connecting):

- Instruct students to begin coloring carefully, making sure to match the numbers in the picture to the correct colors in the key.
- Remind them to also connect the numbered circles on the side in order after they are done coloring.

- Circulate around the classroom to provide assistance and encouragement.

4. Discovery Phase (Observing the Traffic Light):

- Once most students have finished coloring and connecting, guide their attention to the colored traffic light in their picture.
- Ask questions like:
 - "What colors did you use for the traffic light?" (Guide them to identify the colors you assigned to the top, middle, and bottom lights).
 - "In what order are the colors on the traffic light?" (Top to bottom).
 - "Does this look like a traffic light you have seen before?"

5. Reflection (Connecting to Real-World Knowledge):

- Lead a brief discussion about the real-world function of traffic light colors:
 - "What does the red light tell drivers (and sometimes walkers) to do?" (Stop).
 - "What does the yellow light usually mean?" (Slow down, get ready to stop).
 - "What does the green light usually mean?" (Go).

- Briefly connect this back to the idea of making the city roads safer.

Connecting to STEAM:

- **Art:** Coloring and visual representation.
- **Science/Technology/Engineering:** Understanding the basic visual design of a traffic light, a piece of technology used for safety and organization.
- **Math:** Number recognition and following a sequence (connecting the numbers).

Differentiation:

- **Support:** Provide a color key with pre-colored dots next to the numbers for students who need more visual cues. Offer verbal reminders about the color matching.
- **Extension:** Ask students who finish early to draw their own traffic light and label the colors. They could also draw a simple scene of a car stopping at a red light.

3. Explore - Experiencing Traffic Light Signals:

Teacher Role:

- Introduce the card game as a fun and active way to understand the meaning of traffic light colors.
- Clearly explain the rules of the game, demonstrating the actions for each color card.
- Facilitate the game, ensuring all students participate and respond appropriately to the signals.

- Guide students to connect the actions in the game to the behavior of cars and pedestrians at real traffic lights.
- Introduce the "Watch it.." section as an optional resource for further learning.

Expected From Students:

- Actively participate in the card game, performing the correct action when a red, yellow, or green card is shown.
- Follow the rules of the game and respond in a timely manner.
- Make connections between the colors in the game and the signals given by real traffic lights.
- (Optional) Watch the provided video to learn more about how traffic lights work.

Materials Needed:

- Large red, yellow, and green cards (one set for the teacher).
- (Optional) Access to a device with internet and a projector or individual student devices for the "Watch it.." video.
- (Optional) QR code reader app if students are using individual devices.

Step-by-Step Instructions:

1. Introduction to the Game (Engage):

- Say something like: "Now we're going to play a fun game to help us understand what the colors on a traffic light mean! I have some special cards."
- Show the red, yellow, and green cards.

2. Explaining the Game Rules (Explain):

- Clearly explain the action for each color:
 - "When I show the **red** card, everyone **stops** like a car at a red light!" (Demonstrate stopping).
 - "When I show the **yellow** card, get **ready to move**! Like a car when the light is about to turn green!" (Demonstrate being ready).
 - "When I show the **green** card, you can **run, jump, or dance** safely! Like cars can go when the light is green!" (Demonstrate some actions).
- Ensure students understand the actions for each color. You might want to practice one color at a time.

3. **Playing the Game (Explore & Elaborate):**

- Begin showing the cards randomly. Encourage students to react quickly with the correct action.
- Vary the pace of showing the cards.
- You can incorporate simple variations, like having them freeze in their action when you show red again.

4. **Connecting to Real Traffic Lights (Explain):**

- After playing the game for a few minutes, gather the students.
- Ask: "How was our game like the traffic lights we see on the road?" (Guide them to see the connection between the colors and the actions/rules).

- Reinforce the meaning of each color in the context of real traffic.

5. "Watch it.." (Optional Extension):

Video link:

<https://www.youtube.com/watch?v=HhrfOgLjBvg&pp=ygUTdHJhZmZpYyBsaWdodHMgc29uZw%3D%3D>

- Introduce the "Watch it.." section.
- Explain that the video will show them more about how real traffic lights work.
- Provide instructions on how to access the video using the QR code or the YouTube link (if applicable to your classroom setup).
- Watch the video as a class or allow students to watch it individually or in small groups.

4. Assessment - Checking Understanding:

Teacher Role:

- Introduce the "Focus" matching activity as a way to check what students remember about the traffic light colors.
- Provide clear instructions on how to match the colors to their meanings.
- Observe students as they complete the matching activity.

- Introduce the "Practice" activity as a creative way for students to apply their understanding.
- Encourage creativity in their "super cool traffic light" creation.

Expected From Students:

- In the "Focus" section, correctly match the traffic light colors (represented by circles) to their meanings (Stop, Move, Be ready) by drawing lines.
- In the "Practice" section, create their own representation of a traffic light using available materials.

Materials Needed:

- Student books with the "Assessment" section.
- Pencils for the matching activity.
- Various craft materials for the "Practice" activity (e.g., construction paper, markers, pipe cleaners, blocks – be flexible based on what you have available).

Step-by-Step Instructions:

1. Focus - Matching Activity (Evaluate):

- Direct students to the "Focus" section.
- Read aloud the instructions: "Can you guess what each traffic light color means? Let's find the right match! Draw a line from the word to the right circle."
- Ensure students understand they need to connect "Stop" to the red circle, "Move" to the green circle, and "Be ready" to the yellow circle.
- Observe students as they complete the matching.

Differentiation:

- **Support (Focus):** For students who need help with matching, you can point to a color and ask what it means, then guide them to the correct word.

Practice - Creating a Traffic Light" Activity:**Teacher Role:**

- Introduce the DIY traffic light activity as a fun and hands-on way to build their own model.
- Provide clear and simple instructions on how to use the cardboard and stationery to create the traffic light.
- Guide students on the correct shape and color order for the lights.
- Encourage creativity in how they construct their traffic light.

Expected From Students:

- Use cardboard pieces to create the basic shape of a traffic light.
- Use coloring tools (crayons, markers) or colored paper to represent the red, yellow, and green lights in the correct order.
- Assemble their traffic light model.
- Explain the meaning of each color on their traffic light.

Materials Needed:

- Pieces of cardboard (various sizes, recycled boxes work well).
- Scissors (child-safe).
- Glue sticks or child-safe liquid glue.

- Crayons, markers, or colored construction paper (red, yellow, green).
- Pencils.
- Rulers (optional, for drawing straight lines).

Step-by-Step Instructions:

1. Preparation (Before the Activity):

- Gather the cardboard pieces and ensure they are in manageable sizes for young children to work with.
- Have scissors, glue, and coloring materials readily available.
- You might want to pre-cut some simple rectangular cardboard shapes for the main body of the traffic light if time is limited or for students who need extra support with cutting.

2. Building the Traffic Light Shape (Do It):

- Instruct students to take a piece of cardboard and decide on the basic shape for their traffic light (a rectangle standing tall works well).
- Guide them on how to safely use scissors to cut their cardboard into the desired shape. Teacher supervision is crucial during cutting.
- Alternatively, if you pre-cut shapes, provide one to each student.

3. Creating the Lights (Do It):

- Explain that traffic lights have three lights: red on top, then yellow, then green on the bottom.

- Provide red, yellow, and green crayons, markers, or pieces of colored paper.
- **Option 1 (Coloring):** If using crayons or markers, instruct students to draw three circles on their cardboard traffic light in the correct order (red at the top, yellow in the middle, green at the bottom) and color them in.
- **Option 2 (Colored Paper):** If using colored paper, have students cut out three circles (you can provide templates if needed) in red, yellow, and green. Then, guide them to glue the circles onto their cardboard traffic light in the correct top-to-bottom order.

4. **Assembly (Do It):**

- Once the lights are colored or glued on, students can add any other decorative elements they like to their traffic light using markers or additional small pieces of cardboard. For example, they could add a small base so it can stand up.

5. **Sharing and Explaining (Elaborate & Evaluate):**

- Once their traffic lights are complete, have students share their creations with the class.
- Ask each student to point to the red, yellow, and green lights on their model and explain what each color tells cars and people to do. This reinforces their understanding of the traffic light signals.

Differentiation:

- **Support:** Provide pre-cut cardboard shapes and circle templates for students who need help with these fine motor skills.

- **Extension:** Encourage advanced students to think about adding more features to their traffic light, like a button to change the lights or a pedestrian crossing signal (even if they are just pretending).

Another option: to 3D design the traffic light on Tinkercad:

<https://www.youtube.com/watch?v=J5D303ISsuE&pp=ygUhdHJhZmZpYyBsaWdodCBkZXNpZ24gb24gdGlua2VyY2Fk>

Improve & Now I Can...

Content Breakdown of the Image:

- **Top Section:** Text reads: "Now go to the improve part in your EDP paper and add this new idea." There is also a small robot character and a simplified representation of an Engineering Design Process (EDP) paper with sections like "Problem/Purpose," "Improve," etc.
- **Bottom Section:** "Now I can..." with a statement: "Design my traffic light and add it to my city." There is a checkbox next to this statement.

4. Improve - Integrating the New Idea:

Teacher Role:

- Guide students to reflect on the traffic light they designed and how it can be incorporated into the city they previously created (either physically if they built a model, or conceptually if it was a drawing).
- Explain the concept of the "Improve" stage in the Engineering Design Process (EDP) in a simple, age-appropriate manner.

- Encourage students to think about how their traffic light will make their city's roads safer.
- Facilitate a brief sharing session where students can explain where they would place their traffic light and why.

Expected From Students:

- Understand that design is an ongoing process and that we can always make improvements.
- Think about where a traffic light would be most useful in their city.
- (If applicable) Physically place their traffic light model in their city model or draw it into their city drawing.
- Verbally explain how their traffic light contributes to road safety in their city.

Materials Needed:

- Student books with this "Improve" page.
- Students' previously created city models or drawings (from a prior lesson, as referenced by "We've built the city" on the first page).
- The traffic lights they created in the "Practice" activity.
- Their "EDP paper" (this might be a simple worksheet or drawing where they are recording their ideas throughout the project – ensure this is age-appropriate and pre-defined).

Step-by-Step Instructions:

1. Review the Traffic Light Design (Connect):

- Remind students of the traffic lights they just created and the meaning of the colors.

2. Introduce the "Improve" Stage (Explain):

- Say something like: "Great job making your traffic lights! Now, engineers always think about how they can make things even better. This is called 'Improve.' We're going to think about how our traffic lights can make our city even safer."
- Point to the simplified EDP diagram on the page and briefly explain that after we have an idea, we think about how to make it work in our project.

3. Integrating the Traffic Light into the City (Elaborate):

- Ask students to think about their city. "Where in our city do you think a traffic light would be most helpful to make the roads safer?" (Guide them towards intersections or busy areas).
- **If they built a physical city model:** Instruct them to place their traffic light model in the chosen location in their city.
- **If they drew a city:** Have them draw their traffic light into their city drawing in a logical place.

4. Adding to the EDP Paper (Do It):

- Direct students to the "improve" section of their EDP paper.
- Ask them to draw a simple picture of their traffic light in their city or write a sentence about where they placed it and why they think it will make the roads safer. Keep this very simple (e.g., "I put it at the crossing so cars can stop.").

5. Sharing (Optional):

- Have a few students share where they placed their traffic light and why they think it's an improvement for their city's safety.

5. Now I Can... (Self-Assessment):

Teacher Role:

- Guide students to reflect on what they have learned and achieved in this lesson.
- Explain that the "Now I can..." statement is a way for them to check their own understanding.

Expected From Students:

- Read the "Now I can..." statement: "Design my traffic light and add it to my city."
- Reflect on whether they feel they have accomplished this.
- (Optional) Check the box if they feel confident in their ability to design a traffic light and understand its role in their city.

Step-by-Step Instructions:

1. Read the Statement (Engage):

- Read the "Now I can..." statement aloud to the class.

2. Encourage Self-Reflection (Evaluate):

- Ask students to think about everything they did in this lesson: learning about traffic lights, coloring one, building one, and thinking about where it goes in their city.
- Ask: "Did you design your own traffic light?" (Yes, through the color-by-number and the DIY activity).

- Ask: "Did you think about where it would go in our city?"
(Yes, they just placed it or drew it).
- Explain that the checkbox is for them to show if they feel like they can now do these things. It's okay if they don't check it – it helps the teacher know if they need more help.

Objectives:

- 1-Understand that machines can be programmed to perform tasks.
 - 2-Recognize the basic sequence of a traffic light (Red, Yellow, Green).
 - 3-Identify basic programming blocks in a visual environment (Scratch Jr).
 - 4-Create a simple plan for a traffic light sequence using code blocks.
 - 5-Integrate a model traffic light into a representation of a city.
- ☐ Understand that technology can be used to improve community safety.

Challenge Questions of the Lesson

- 1-Why do traffic lights have red, yellow, and green in that order?
- 2-What's one simple way to make a traffic light even better or "smarter"?
- 3- What is coding?

SEL relation

- 1-Self-Awareness: Reflect** on understanding of rules and safety.
- 2-Self-Management:** Follow instructions for activities.
- 3-Social Awareness:** Consider how traffic lights ensure safety for everyone.
- 4-Relationship Skills:** Share ideas and collaborate (if applicable).
- 5-Responsible Decision-Making:** Understand traffic lights help with safe choices.

1. Engage:

Teacher Role:

- Introduce the exciting next step of making their model traffic light work using technology, connecting it to their prior experience of building the city and the traffic light outline.
- Incorporate visuals of simple robots or automated toys and ask open-ended questions to spark curiosity about how machines follow instructions.
- Encourage students to make predictions about what "programming" might mean in the context of making their traffic light change colors.
- Facilitate a supportive discussion where students can share their initial ideas about how to tell a machine what to do.
- Provide gentle guidance by relating the concept of programming to familiar sequences they follow in their daily routines.

Expected From Students:

- Share their ideas about how toys or machines work and follow instructions.
- Look at visuals of robots or automated toys with curiosity.
- Offer their initial thoughts on what it means to "program" something like a traffic light.
- Listen respectfully to their classmates' ideas about making the traffic light work.
- Begin to connect the idea of giving instructions to a machine with the traffic light changing colors.

Strategies for implementation

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- **Preview with a Hook:**
 - **Strategy:** Begin with a question about familiar automated objects.
- **Implementation:**
 - Ask, “Think about a toy that does something by itself, like a car that moves when you press a button. How does it know what to do?”
 - Discuss briefly, allowing students to share their ideas (e.g., batteries, switches, maybe even something inside telling it what to do).
- **Make Predictions:**
 - **Strategy:** Introduce Mrs. Sara's statement about making the traffic light work.
 - **Implementation:**
 - Say, "Remember Adam said they were almost done? Mrs. Sara said, 'Yes, kids. Let's make this traffic light work!' How do you think we can make our traffic light change its colors all by itself?"

- Ask students, “What do you think we might need to tell the traffic light to make it turn red, then yellow, then green?”

- **Set the Scene with a Visual:**

- **Strategy:** Use a simple visual of a robot or a sequence of lights changing.
- **Implementation:**
 - Show a picture of a simple robot or a drawing showing a traffic light changing colors in order.
 - Ask, “What is this robot doing? How does it know what to do next?” or “What do you notice about the lights in this picture? Do they change in a special order?”

- **Think-Pair-Share (Simplified):**

- **Strategy:** Simple partner talk about giving instructions.
 - **Implementation:**
 - Ask, "Think about telling a friend how to build a tall tower with blocks. What would you tell them to do first? What would be the next step?"
 - Tell them to talk to their friend next to them about giving instructions.
 - Ask a few students to share one step they told their friend.
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2. Let's Think - Identifying Traffic Light Sequences:

Teacher Role:

- Introduce the "Let's Think" activity as a puzzle to reinforce the correct order of traffic light colors.
- Explain that each row shows a pattern of traffic light colors (Red, Yellow, Green), but one color is missing in each pattern.
- Guide students to recall the standard sequence of traffic lights to determine the missing color in each row.
- Facilitate a discussion after the activity to confirm the missing colors and reiterate the importance of the Red-Yellow-Green order for road safety.

Expected From Students:

- Examine each numbered pattern of colored circles (Red, Yellow, Green).
- Remember the correct sequence in which traffic lights usually change colors: Red, then Yellow, then Green.
- Identify the missing color in each sequence based on their knowledge of the traffic light order.
- Circle the colored circle (Red, Yellow, or Green) that represents the missing color for each pattern.
- Participate in a discussion to review the correct answers and explain their reasoning based on the standard traffic light sequence.

Materials Needed:

- Student books with the colored "Let's Think" activity page.

- Pencils for circling the answers.

Step-by-Step Instructions:

1. Introduction (Engage):

- Say something like: "Great job learning about traffic lights! Now, let's do a color puzzle. Each row here shows a pattern of traffic light colors – Red, Yellow, and Green – but one color is missing in each line. We need to be super detectives and find the missing color!"
- Briefly remind students of the correct order: "What color is on top of a real traffic light? (Red) What comes next? (Yellow) And what's the last color before it changes again? (Green)."

2. "Do It" Activity (Solving the Patterns):

- Instruct students to look at the first row. "What colors do you see? Think about the order: Red, Yellow, Green. What color do you think is missing to keep the pattern going correctly?"
- Guide them through the first one if needed, then encourage them to solve the remaining patterns on their own. Emphasize that they should circle the colored circle (Red, Yellow, or Green) below the pattern that they think is missing.

3. Discovery Phase (Reviewing the Answers):

- Once students have had time to complete the activity, go through each pattern one by one.

- Ask: "For number 1, what color did you circle? Why did you choose the [Color] circle? How does it fit into the Red, Yellow, Green order?" Encourage them to explain their reasoning.
 - Confirm the correct missing color for each pattern:
 - 1. Missing: red
 - 2. Missing: red
 - 3. Missing: yellow
 - 4. Missing: red
 - 5. Missing: yellow
 - Briefly discuss why the traffic light follows this specific sequence for safety.
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3. Explore - Programming the Traffic Light Sequence (ScratchJr - Block Cutting Activity):

Teacher Role:

- Explain that students will now plan the code for their smart traffic light by choosing and arranging paper cutouts of ScratchJr blocks.
- Provide the pre-printed sheet of ScratchJr blocks and guide students on how to carefully cut them out.

- Facilitate a discussion about which blocks might be needed to make the traffic light change colors in the correct sequence (Red, Yellow, Green) with pauses in between.
- Encourage students to work individually or in small groups to arrange their chosen blocks in the logical order.
- Circulate and provide support, helping students understand the function of different blocks and the flow of the program.

Expected From Students:

- Carefully cut out the individual ScratchJr block images from the provided sheet.
- Identify and select blocks related to:
 - **Starting the program:** (e.g., the green flag block, or the "start on tap" block if they recall it).
 - **Changing the appearance/color of the traffic light character:** (Blocks with color palettes or names like "set color"). *Teacher Note: Since the visual doesn't show specific color blocks, guide them to imagine blocks that would change the character's color.*
 - **Introducing a pause or delay between color changes:** (The "wait" or clock block).
 - **Repeating the sequence:** (The repeat block, if they grasp the concept).
- Arrange the chosen blocks in a logical sequence that they believe would make the traffic light change colors in the order: Red, then Yellow, then Green, with a pause after each color.

- Be able to explain their chosen sequence of blocks and why they think it would work.

Materials Needed:

- Teacher-prepared printable sheet containing a screenshot or clear representation of relevant ScratchJr blocks (including at least one start block, appearance/color change blocks - even if generic color palette icons, and a wait/speed control block, and optionally a repeat block). *See "Teacher Preparation" below for guidance on creating this sheet.*
- Scissors (child-safe).
- Large sheets of paper or designated workspace for students to arrange their blocks.
- Glue sticks (optional, if you want them to temporarily stick their arrangements down).

Step-by-Step Instructions:

1. Teacher Preparation (Crucial):

- **Take a screenshot of the ScratchJr interface, focusing on the "Looks" (Appearance) category and the "Control" category.** Ensure the blocks are clear and large enough to be easily cut out.
- **Alternatively, create a simplified visual representation of the necessary blocks.** You could draw rectangles representing blocks with simple icons or labels (e.g., a green flag for start, a paint palette for color, a clock for wait, a looping arrow for repeat).

- **Print enough copies of this block sheet for each student or pair of students.**

2. Introduction (Connect to Digital Programming):

- Say something like: "Now, we're going to plan how to tell the computer to make our traffic light work in ScratchJr! These are what the instructions look like – they are like puzzle pieces called blocks."
- Distribute the printed sheets of ScratchJr blocks and scissors.
- Instruct students to carefully cut out each block.

3. Exploring the Blocks (Explore):

- Briefly review the different types of blocks they have cut out. Focus on:
 - **Start Block:** "This block tells the program when to begin."
 - **Appearance/Color Blocks:** "These blocks will help us change the color of our traffic light. Let's pretend these color palettes will let us pick red, then yellow, then green."
 - **Wait Block:** "This block tells the program to wait for a little while before doing the next thing."
 - **Repeat Block (Optional):** "This block can make the program do the same thing over and over again."

4. "Do It" Activity (Arranging the Blocks):

- Explain the task: "Your challenge is to choose the blocks you think we need and arrange them in the right order to make our traffic light turn red, wait, then turn yellow, wait, then turn green, and maybe even do it again!"
- Encourage students to work individually or in small groups to arrange the blocks on their paper in the sequence they think is correct.
- Prompt their thinking: "What block should come first? What color should it change to? What should happen after it turns red? What color comes next?"

5. Sharing and Discussion (Elaborate):

- Have students share their block arrangements with the class.
- Ask them to explain why they chose that order and what they think each block will do.
- Guide the discussion towards the correct logical sequence:
Start -> Change to Red -> Wait -> Change to Yellow -> Wait -> Change to Green -> (Optional: Repeat).

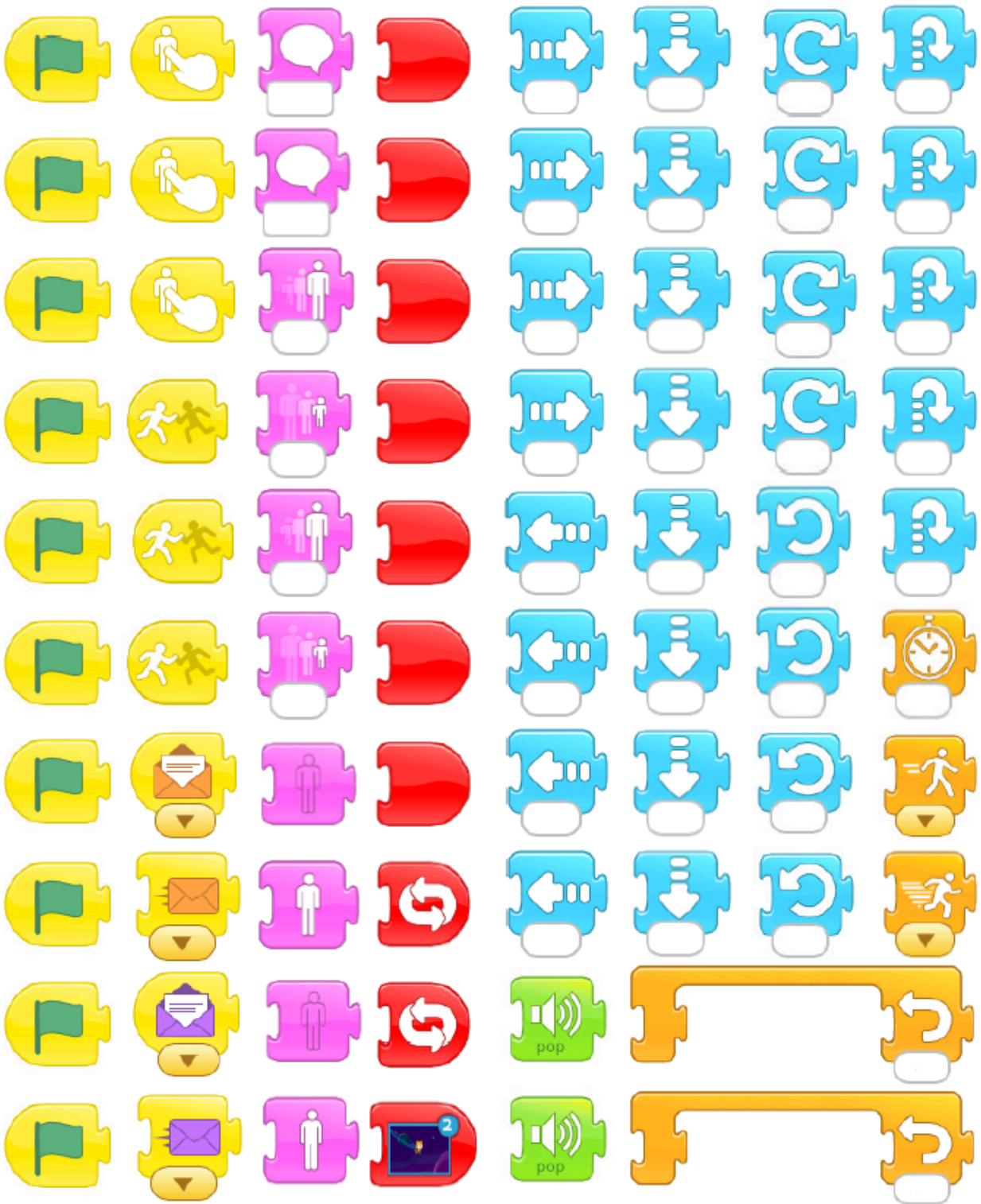
Connecting to STEAM:

- **Technology:** Understanding that visual blocks represent code instructions for a computer.
- **Engineering:** Planning a sequence of actions to achieve a desired outcome (making the traffic light work).
- **Computational Thinking:** Developing algorithmic thinking by sequencing steps.

Differentiation:

- **Support:** Provide a suggested starting block and guide students through the first few steps of the sequence.
 - **Extension:** Challenge advanced students to incorporate a repeating loop in their block arrangement to make the traffic light cycle continuously.
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Resources teacher will need :





<https://scratchjr.fun.com/traffic-light-sequence-with-messages/>

Links for the pictures:

<https://www.scratchjr.org/assessments/block-labels.png>

https://images.twinkl.co.uk/tw1n/image/private/t_630_eco/image_rep_o/e8/5d/t-i-028-scratchjr-blocks_ver_1.jpg

4. Watch it.. - Exploring What Coding Is:

Teacher Role:

- Introduce the "Watch it.." video as an opportunity for students to learn more about the concept of coding in a fun and accessible way.
- Explain that coding is like giving instructions to a computer to make things happen, similar to how they are planning with the ScratchJr blocks.
- Provide clear instructions on how students can access the video using the QR code or the YouTube link (depending on your classroom setup and available technology).
- Facilitate a brief discussion after watching the video to reinforce the main ideas about what coding is.

Expected From Students:

- Understand that coding is a way to tell computers what to do.
- Recognize that coding involves giving instructions in a specific order.
- Make connections between the concept of coding explained in the video and the block-based programming they will be doing in ScratchJr.
- (Optional) Share any interesting things they learned about coding from the video.

Materials Needed:

- Student books with the "Watch it.." section and QR code/YouTube link.

- Devices with internet access (tablets, computers, or a projector-connected device).
- QR code reader app (if students are using tablets to scan the code).

Step-by-Step Instructions:

1. Introduction (Connect to the "Explore" Activity):

- Say something like: "We've been thinking about how to tell the computer what to do using our block plan. Now, let's watch a short video that explains what this is all about – it's called coding!"
- Explain that the video will help them understand what coding means and how it works.

2. Accessing the Video:

- Provide clear instructions on how to watch the video:
 - **If using QR codes:** "If you have a tablet with a QR code reader, you can open the app and point it at this square (show the QR code). It will take you right to the video."
 - **If using a YouTube link:** "If you are using a computer or tablet, you can type in this link (show the YouTube link clearly) into the internet browser to watch the video."
- Ensure all students can access the video successfully or plan to watch it as a whole class.

3. Watching the Video:

- Play the video for the students. If watching as a class, ensure it is displayed clearly for everyone to see.

4. **Post-Viewing Discussion (Reflect and Connect):**

- After the video, have a brief discussion using simple questions:
 - "What did the video say that coding is?" (Guide them towards the idea of giving instructions to a computer).
 - "Did you see anything in the video that looked a little bit like the blocks we are going to use?" (If the video shows visual programming).
 - "Why do people learn to code?" (Guide towards ideas like making games, robots work, etc. - keep it simple).
 - "Does coding sound like something that needs to be done in a specific order?" (Yes, connect it back to their block arrangement).

Differentiation:

- **Support:** If some students have difficulty accessing the video, consider showing it as a whole class. Pause at key points to explain concepts further.
- **Extension:** Ask students who are particularly interested to share anything else they know or have heard about coding.

5. Assessment - Checking Understanding (Focus):

Teacher Role:

- Introduce the "Focus" section as a way to check what students have learned about algorithms, computer code, and sequences.
- Read each question aloud clearly to the students.
- Ensure students understand the multiple-choice format and how to select their answers (circling the letter).
- Circulate to offer support if needed, but encourage students to answer based on their own understanding.
- Review the correct answers as a class after students have completed this section, explaining the reasoning behind each.

Expected From Students:

- Listen attentively as the teacher reads each question.
- Think critically about the concepts of algorithms, computer code, and sequences based on the lesson activities and discussions.
- Select the best answer for each multiple-choice question by circling the corresponding letter.
- Participate in the review of the answers and listen to the teacher's explanations.

Materials Needed:

- Student books with the "Assessment" page.
- Pencils for circling answers.

Step-by-Step Instructions:

1. Introduction to "Focus":

- Say something like: "Now, let's see what we've learned about how computers think! This 'Focus' section has some questions for you to answer. Listen carefully as I read each one."

2. Administering the Questions:

- Read question 1: "What is an algorithm similar to?"
 - A) a song
 - B) a recipe
 - C) a picture
 - D) a toy."

(Correct Answer: B) -

Teacher Note: Explain that an algorithm is like a set of step-by-step instructions, just like a recipe tells you how to make food.
- Read question 2: "What do we call the language we use to talk to computers?"
 - A) Animal language
 - B) Computer code
 - C) Sign language
 - D) Body language.

(Correct Answer: B) –
- *Teacher Note: Remind them that the blocks in ScratchJr are a type of computer code.*

- Read question 3: "What does a sequence mean in coding?"
- A) a group of numbers
- B) a list of colors
- C) steps in order
- D) a bunch of words.
- (Correct Answer: C) –
- *Teacher Note: Emphasize that in coding, the order of the instructions (the sequence of blocks) is very important.*

3. Student Response:

- Instruct students to circle the letter of the answer they think is correct for each question. Allow sufficient time for them to consider their choices.

4. Review and Discussion:

- Once everyone has finished, go through each question and reveal the correct answer.
- For each question, briefly explain *why* the correct answer is right and why the other options are not the best fit.

6. Practice - Finishing the Scratch Jr Game:

Teacher Role:

- Introduce the "Practice" section as an opportunity for students to apply their understanding of coding sequences in ScratchJr.
- Explain that they will be working on a simple game involving a car and traffic light colors.

- Provide clear instructions on the goal of the game (likely to program the traffic light colors to change in the correct sequence so the car knows when to stop and go).
- Guide students to use the ScratchJr blocks they explored earlier to build the code for the traffic light.
- Offer support and guidance as students work on their ScratchJr projects.

Expected From Students:

- Open the ScratchJr app on their tablets or computers.
- Locate the pre-started game project (if you have set one up) or create a new project with a car and colored circles (acting as the traffic light).
- Use the ScratchJr blocks to program the colored circles to change in the sequence: Green (for the car to move), then Yellow (get ready), then Red (for the car to stop), with appropriate pauses.
- Test their code by running the program and observing if the car reacts correctly to the changing "traffic light."
- Debug their code if it doesn't work as expected.

Materials Needed:

- Tablets or computers with the ScratchJr app installed.
- (Optional) Pre-started ScratchJr project with a car sprite and colored circle sprites positioned as a traffic light.
- (Optional) Step-by-step visual guides or instructions for basic ScratchJr block usage for this task.

- Video link : <https://www.youtube.com/watch?v=hmqOcswc8vs>

Step-by-Step Instructions:

1. Introduction to "Practice":

- Say something like: "Now it's time to be real coders! In this 'Practice' section, we're going to finish a game in ScratchJr where we make our traffic light work to help a car move safely."

2. Setting up the Game (if needed):

- If you have a pre-started project, instruct students on how to open it.
- If not, guide them on how to create a new project, add a car sprite, and draw or add three colored circle sprites (red, yellow, green) and position them like a traffic light.

3. Programming the Traffic Light:

- Remind students of the block arrangements they planned earlier.
- Guide them on how to use the ScratchJr interface to drag and connect the appropriate blocks to the colored circle sprites to create the traffic light sequence:
 - Start the sequence (e.g., "Start on Tap" for one of the light sprites).

- Make the green light appear in some way (e.g., "Show" block if it was hidden initially, or just have it as the starting state).
- Add a "Wait" block.
- Make the green light disappear or change its behavior, and make the yellow light appear.
- Add another "Wait" block.
- Make the yellow light disappear and the red light appear.
- Add a final "Wait" block.
- (Optional: Add a "Repeat" block to make the sequence loop).

4. Programming the Car (Basic Interaction):

- You can also guide students to add very basic code to the car sprite, such as moving forward when the green light is "on" (this might be a visual cue they need to observe).

5. Testing and Debugging:

- Encourage students to tap the "Start" block and observe if their traffic light sequence works as intended and if the car's movement makes sense with the lights.

- Remind them that if it doesn't work right away, they need to look at their code and try to find the mistake (debugging).

6. Sharing (Optional):

- If time allows, have students share their working games with the class and explain how their code makes the traffic light and car behave.
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7. Introducing 7-Segment Displays:

Teacher Role:

- Read aloud Laila's speech bubble and introduce the concept of a 7-segment display as another way traffic lights (and other devices) can show information, like time.
- Show the illustration of the 7-segment display and explain how the different segments can light up to form numbers.
- Connect this to the idea of "smart" traffic lights that might show countdowns or other information.

Expected From Students:

- Listen to the information about 7-segment displays.
- Observe the illustration and understand that it can show numbers.
- Begin to associate 7-segment displays with technology that provides information.

Materials Needed:

- Student books with this page.
- (Optional) A real 7-segment display if available.

Step-by-Step Instructions:

1. Introduce 7-Segment Displays:

- Read Laila's speech bubble with enthusiasm.
- Say something like: "Wow, Laila noticed something cool! Some traffic lights, especially 'smart' ones, can even show us numbers using something called a 7-segment display. Look at this picture."

2. Explain the Display:

- Point to the illustration of the 7-segment display.
- Explain that it has seven little lines (segments) that can light up.
- Show how different segments light up to make the number "8" in the example. You can trace the segments with your finger.
- Explain that by lighting up different combinations of these seven lines, the display can show all the numbers from 0 to 9, and sometimes even letters.

3. Connecting to Smart Traffic Lights:

- Remind students that they were thinking about "smart" traffic lights.
 - Ask: "How could a 7-segment display be useful on a smart traffic light?" (Guide them towards ideas like showing how many seconds are left before the light changes).
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Showcase

Teacher Role:

- Facilitate sharing of DIY traffic lights integrated into their cities.

- Provide positive feedback.
- Guide self-reflection on "Now I can..." statements.

Expected From Students:

- Show their DIY traffic light placed within their city model or drawing.
- Explain how their DIY traffic light helps make their city safer.
- Check boxes for "Now I can..." statements they understand.

Materials Needed:

- Students' DIY traffic light models.
- Students' city models/drawings.
- Student books.

Step-by-Step Instructions:

1. **Sharing:** Students show their DIY traffic lights within their cities, explaining its role in safety. Give positive feedback.
 2. **Self-Reflection:** Read "Now I can..." statements. Students check boxes if they agree. Briefly discuss.
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Create your Story

Teacher Role:

- Introduce the activity as a creative way to summarize the lesson.
- Encourage students to use the four frames to tell a story about their smart traffic light and its impact on the city.
- Remind them to include key elements from the lesson.

Expected From Students:

- Use the four frames to draw a sequence of events.
- Tell a simple story about their smart traffic light.
- Include the traffic light, the city, and the idea of safety.

Materials Needed:

- Student books with the "Create your Story" page.
- Pencils and coloring materials.

Step-by-Step Instructions:

1. **Introduce:** Explain students will draw a four-part story about their smart traffic light and the city.
2. **Create:** Students draw a sequence in the four frames. Encourage including the traffic light, city, and safety.
3. **Share (Optional):** Have students briefly share their stories.

